The Exact Limiting Distribution of the Dickey–Fuller t-Ratio

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Abstract

The limiting distribution of the Dickey–Fuller t-ratio, the reference distribution of the most widely used test in time series econometrics, has been tabulated by simulation since 1979. We obtain its density and distribution function in closed form up to a single integral, for the regression without deterministic terms, with an intercept, and with an intercept and a linear trend, under local alternatives $\phi = 1 + c/n$ on either side of the unit root, for the t-ratio centered at the unit root or at any local null value, and with a zero, a fixed or a stationary initial observation of the order of $\sqrt{n}$. The joint moment generating function of the numerator and the denominator of the limit is a Fredholm determinant that is a polynomial of degree at most two in the argument of the numerator, because the numerator is a quadratic form of rank at most two in a few Gaussian functionals of the path, and the initial observation enters it through a closed-form quadratic term. An inversion formula for ratios of the form $X/\sqrt{Y}$ then gives the density and the distribution function as single integrals of special functions along a line in the complex plane; reversing the order of the two integrations gives a second representation, in which the inner integral runs along saddle-point paths, and the two check each other. The integrals are evaluated to eleven or twelve digits in a fraction of a second, and the moments of the limits are one-dimensional integrals of the moment generating function at real arguments. The results reproduce the response-surface critical values of MacKinnon, the moments of Nabeya, the closed forms of Abadir and the

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simulated local powers in the literature, and make the limiting power of the three tests, with a zero, a fixed or a stationary start, exact.

Keywords: unit root, Dickey–Fuller test, t-ratio, local-to-unity asymptotics, Fredholm determinant, inversion formula, parabolic cylinder function, saddle point contour

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1. Introduction

The Dickey–Fuller t-ratio is the least squares t-statistic for $\phi = 1$ in the first-order autoregression $y_t = \alpha + \phi y_{t-1} + \sigma e_t$, computed from the regression of y_t on y_{t-1} (the statistic $\hat{\tau}$), on a constant and y_{t-1} ($\hat{\tau}_\mu$), or on a constant, a linear trend and y_{t-1} ($\hat{\tau}_\tau$) (Dickey and Fuller, 1979, 1981; Fuller, 1996). Under the unit root its limiting distribution is a functional of Brownian motion (Dickey and Fuller, 1979; Chan and Wei, 1987; Phillips, 1987), and under the local alternatives $\phi = 1 + c/n$ a functional of an Ornstein–Uhlenbeck process. Every critical value in use—the tables of Fuller (1996), the response surfaces of MacKinnon (1996, 2010) and Cheung and Lai (1995), and the tables built into current software—comes from simulation, and so do the local power curves of the three tests (Elliott, 1999; Müller and Elliott, 2003; Harvey et al., 2009).

The reason is that the t-ratio is not a ratio of quadratic forms. For the coefficient statistic $n(\hat{\phi} - 1)$, whose limit is the ratio of two quadratic forms in the path, the joint characteristic function of numerator and denominator is a Fredholm determinant (Nabeya and Tanaka, 1990a), and the distribution function follows by the inversion formula of Imhof (1961) and Gil-Pelaez (1951) as a one-dimensional integral; Nabeya and Tanaka (1990b) and Nabeya and Sørensen (1994) tabulated the limiting distributions of the coefficient statistics in this way, under local alternatives and with a nonzero initial value, a drift and a trend, and Tanaka (1990) and Perron (1991) treated the fixed and the stationary initial observation. The limit of the t-ratio is the ratio of a quadratic form to the square root of another, which this approach does not reach (Tanaka, 2017, Section 8.4). What is known exactly is partial. Abadir (1992) computed the limiting density and distribution function of $\hat{\tau}$ numerically, from a formula with a double sum and an integral, and Abadir (1995) derived a convergent series for its

lower tail and single integrals of parabolic cylinder functions for its upper tail, for $\hat{\tau}$ at the unit root with a zero start (Abadir (1993) did the same for the coefficient statistic). Larsson (1995) gave an exact formula for the upper tail of the limiting distribution of $\hat{\tau}$ under local alternatives and saddlepoint approximations to both tails, and Larsson (1998) saddlepoint approximations to its lower tail; Hitomi et al. (2022) wrote the limiting distributions of the coefficient and t statistics without an intercept, with a zero or a fixed initial value, as double integrals of the joint density of a squared Bessel process and its time integral, whose integrands are triple sums of parabolic cylinder functions. Nabeya (1999) computed the first six moments of the three limiting null distributions, by the formula of Sawa (1978), and Larsson (1997) the second moment of $\hat{\tau}_\mu$. For $\hat{\tau}_\mu$ and $\hat{\tau}_\tau$, under the null and under local alternatives, no exact result for the distribution has been available, and the effect of the initial observation on the local power of the three t-tests has been studied by simulating the limiting functionals (Elliott, 1999; Müller and Elliott, 2003; Harvey et al., 2009).

This paper gives the exact limiting density and distribution function of the three t-ratios, under the unit root and under local alternatives on either side of it, for the statistic centered at the unit root or at any local null value $\phi_0 = 1+c_0/n$, and with a zero, a fixed or a stationary initial observation of the order of $\sqrt{n}$, as single integrals of closed-form integrands, and evaluates them to eleven or twelve digits in a fraction of a second. The representation rests on two facts. First, the limiting numerator and denominator of the t-ratio, $U = \int_0^1 \tilde{J}\mathrm{d}J$ and $V = \int_0^1 \tilde{J}^2\mathrm{d}r$ in the notation of Section 2, have a joint moment generating function that is a Fredholm determinant (Nabeya and Tanaka, 1990a; Nabeya and Sørensen, 1994), and this determinant is a polynomial of degree one or two in the argument of U, because U is a quadratic form of rank one or two in a few Gaussian functionals of the path, $J(0)$, $J(1)$, $\int_0^1 J\mathrm{d}r$ and $\int_0^1 rJ\mathrm{d}r$; a fixed initial observation adds to its logarithm a closed-form term that is quadratic in the start, and a stationary start, averaged over, leaves a determinant of the same degree. Second, the distribution of a ratio $X/\sqrt{Y}$ is a double inversion of the joint moment generating function of (X, Y) with an explicit kernel; the polynomial structure makes the inversion in the argument of U explicit, a scaled χ^2_1 density for $\hat{\tau}$ and the density of a difference of two scaled χ^2_1 variables, a Bessel function K_0, for $\hat{\tau}_\mu$, $\hat{\tau}_\tau$ and the three

stationary cases, and what remains is a single integral over the argument of V of parabolic cylinder functions and of Gaussian transforms of K_0 along a line in the complex plane. The integrand decays only like $\exp(-\frac{1}{2}|\operatorname{Im} t|^{1/2})$ along the line and, in the lower tail, oscillates with a period proportional to the square of the argument; a partition of the integrand into an analytic part and an oscillating part, each integrated by a rule suited to it, gives twelve digits with a few hundred nodes. Reversing the order of the two integrations gives a second representation of the same double integral, in which the outer integral runs over the value of V and the inner one along a path through a saddle point, on which the integrand decays fast whatever the argument; it serves as an independent check of the first, which covers a fixed nonzero start as well, the inversion in the argument of U being then a one-dimensional integral rather than a closed form in the regressions with deterministic terms, and it takes over far in the tails, where the error of the first, absolute rather than relative, shows. The moments of the limits need no inversion at all: a gamma integral makes them one-dimensional integrals of the moment generating function at real arguments.

The principle is more general than the problem. Whenever a statistic is a ratio $X/\sqrt{Y}$ in which $X+a$ is a quadratic form of low rank r in a Gaussian vector and Y a quadratic form in the same vector, the joint moment generating function is e^{-as} times the inverse square root of a polynomial of degree r in s whose coefficients depend on t alone, because under the tilt e^{tY} the vector remains Gaussian and $X+a$ remains a quadratic form of rank r. The inversion in s is then the density of a weighted sum of r squared normals, in closed form for $r \leq 2$, and the distribution of the ratio is a single integral in t of that density against an explicit Gaussian kernel. The Dickey–Fuller limits are the case $r \leq 2$ with $a = \frac{1}{2}$, and so are the limits of the unit root tests with a drift or a trend in the data-generating process. The limit of the t-ratio in a predictive regression with a near-integrated regressor is the same ratio, scaled by the correlation of the innovations, convolved with an independent normal, since the part of its numerator that is orthogonal to the regressor's innovations is conditionally Gaussian given the path.

The results are checked against everything that is known. The distribution function reproduces the probability $\mathbb{P}(\hat{\tau} \leq 0)$, which is known in closed form, to 10^{-13} for c between

-30 and 20, the series and the integrals of Abadir (1995) to 10^{-15}, the probability $\mathbb{P}(\hat{\tau}_\mu \leq 0)$ computed by Bao and Kan (2026) from the distribution of the coefficient statistic to its eight digits, and the response-surface critical values of MacKinnon (1996, 2010) to their stated accuracy; the two representations agree with each other to 10^{-11} over the six cases, both sides of the unit root and the local null values; the moments reproduce the values of Nabeya (1999) and Larsson (1997); the local powers reproduce the simulations of Larsson (1995) and Elliott (1999) within their sampling error; and two million simulated paths of the limiting functionals agree with the distribution functions within their standard errors. The limiting power of the three tests with a fixed or a stationary start, studied so far by simulation, is then exact: the start raises power, most for the test without deterministic terms, and lowers it slightly for the test with a trend close to the unit root. The finite-sample distribution of the same statistics, which the limits approximate, is the subject of the companion paper (Bao et al., 2026), whose exact finite-sample powers we set beside the limits.

Section 2 sets out the models and their limits, Section 3 the joint moment generating function, its structure and the initial observation, Section 4 the inversion formulas and the single integrals for the density and the distribution function, Section 5 the second representation, Section 6 the moments, Section 7 the evaluation, and Section 8 the numerical results. Proofs are in Appendix A and the details of the quadrature in Appendix B. The MATLAB programs that produce every number in the paper accompany it, together with implementations of both representations in Julia and in C, the latter with a MATLAB interface.

2. The *t*-ratios and their limits

Consider the first-order autoregression

$$y_t = \alpha + \delta t + \phi y_{t-1} + \sigma e_t, \qquad t = 1, \ldots, n, \tag{1}$$

with $e_1, \ldots, e_n$ independent $N(0,1)$ and $\sigma > 0$. Following Bao and Kan (2026) we label the cases by the initial observation, Deterministic, that is fixed, or Stationary, and by the deterministic terms, intercept Known to be zero, intercept Unknown, or intercept and Trend

unknown. In case DK ($\alpha = \delta = 0$ known, y_0 fixed) ϕ is estimated by the regression of y_t on y_{t-1}, in case DU ($\delta = 0$ known) by the regression on a constant and y_{t-1}, and in case DT by the regression on a constant, t and y_{t-1}. Let $\hat{\tau}$ be the least squares t-ratio for the hypothesis $\phi = 1$, that is $\hat{\phi} - 1$ divided by its standard error: the Dickey–Fuller statistics $\hat{\tau}$, $\hat{\tau}_\mu$ and $\hat{\tau}_\tau$. More generally let $\hat{\tau}_0$ be the t-ratio for the hypothesis $\phi = \phi_0$ with $\phi_0 = 1 + c_0/n$, the Dickey–Fuller statistic at $c_0 = 0$ and the t-ratio centered at the true value at $c_0 = c$, whose distribution is needed when a confidence interval for ϕ is obtained by inverting the test. Cases SK, SU and ST are cases DK, DU and DT with $|\phi| < 1$ and the initial observation drawn from the stationary distribution of the process.

Let $\phi = 1 + c/n$ with c fixed as $n \to \infty$, no drift, and $y_0 = \sigma\sqrt{n}\,\gamma_0$. Let W be a standard Brownian motion on $[0, 1]$ and

$$J(r) = \gamma_0 e^{cr} + \int_0^r e^{c(r-u)} \mathrm{d}W(u), \qquad \mathrm{d}J = cJ\mathrm{d}r + \mathrm{d}W, \tag{2}$$

the Ornstein–Uhlenbeck process started at γ_0, where γ_0 is a constant in the fixed-start cases ($\gamma_0 = 0$ is the standard setting, in which the start is negligible against $\sqrt{n}$) and $\gamma_0 \sim N\{0, -1/(2c)\}$ independent of W, the limit of the stationary distribution of $y_0/(\sigma\sqrt{n})$, in the stationary cases, which require $c < 0$. Following Müller and Elliott (2003) we also write $\gamma_0 = \xi/\sqrt{-2c}$, so that ξ is the initial deviation in units of the stationary standard deviation and the stationary start is $\xi \sim N(0, 1)$. Let $\tilde{J}$ be J (DK, SK), J less its mean $\int_0^1 J\mathrm{d}r$ (DU, SU), or J less its projection on $(1, r)$ in $L^2[0, 1]$ (DT, ST). Then, as $n \to \infty$ (Phillips, 1987; Chan and Wei, 1987; Nabeya and Sørensen, 1994),

$$\hat{\tau}_0 \Rightarrow \tau = \frac{U - c_0 V}{\sqrt{V}}, \qquad U = \int_0^1 \tilde{J}(r)\mathrm{d}J(r), \qquad V = \int_0^1 \tilde{J}(r)^2\mathrm{d}r, \tag{3}$$

since $\hat{\tau}_0 = \hat{\tau} - c_0/(n\,\mathrm{se})$ and $n\,\mathrm{se} \to V^{-1/2}$, where se is the standard error of $\hat{\phi}$. Here U, the limit of $n(\hat{\phi} - 1)$ times the limit of $n^2\mathrm{se}^2$, is defined by the formula $\int_0^1 J\mathrm{d}W + c\int_0^1 J^2\mathrm{d}r$ less the projection terms, since $\tilde{J}$ depends on the whole path and the stochastic integral is not an Itô integral of $\tilde{J}$. In cases DU, DT, SU and ST the regression removes γ_0 itself and only $\gamma_0(e^{cr} - 1)$ remains, which vanishes at $c = 0$: the null limits do not depend on the start, and the start matters only under the alternative. We write $\mathcal{F}$ and f for the distribution function and the density of the limit in question.

Two identities organize what follows. By Itô's formula, $U+\frac{1}{2}$ is a quadratic form of rank one or two in a few Gaussian functionals of the path. With $\bar{J}=\int_0^1 J\mathrm{d}r$ and $K=\int_0^1 rJ\mathrm{d}r$,

$$U+\frac{1}{2}=\frac{J(1)^2-J(0)^2}{2}-L, \qquad L=\begin{cases} 0 & (\mathsf{K}),\\ \{J(1)-J(0)\}\bar{J} & (\mathsf{U}),\\ \{J(1)-J(0)\}(4\bar{J}-6K) & \\ \quad +6\{J(1)-\bar{J}\}(2K-\bar{J}) & (\mathsf{T}), \end{cases} \tag{4}$$

which for a zero start reduces to $\frac{1}{2}J(1)^2$, $\frac{1}{2}J(1)^2-J(1)\bar{J}$ and $\frac{1}{2}J(1)^2-J(1)\bar{J}-12\{\frac{1}{2}J(1)-\bar{J}\}(K-\frac{1}{2}\bar{J})$. The rank is one in case DK and two otherwise; in the cases with deterministic terms U is invariant to $J\mapsto J+a$ (U) or $J\mapsto J+a+br$ (T), as the regression is, so that $U+\frac{1}{2}$ is a quadratic form in two invariant linear combinations of these functionals, $J(1)-J(0)$ and $J(1)+J(0)-2\bar{J}$ in case U. And given V the event $\{\tau\le x\}=\{U\le x\sqrt{V}+c_0V\}$ is linear in U. The first identity makes the joint moment generating function of (U,V) a polynomial in the argument of U (Section 3); the second is what the inversion formulas of Sections 4 and 5 exploit. We write $R=U+\frac{1}{2}$ where the quadratic form is in view, and $a(v)=\frac{1}{2}+x\sqrt{v}+c_0v$, so that $\{\tau\le x\}=\{R\le a(V)\}$.

3. The joint moment generating function

Let $M(s,t)=\mathbb{E}\exp(sU+tV)$, which is finite on a tube $\{(\mathrm{Re}\,s,\mathrm{Re}\,t)\in\mathcal{D}\}$ with $\mathcal{D}$ a convex neighborhood of the origin. With a zero start it is known in closed form: the Fredholm determinants of Nabeya and Tanaka (1990a, Theorem 4) for their Models 1, 2 and 4 give it in the three cases, after their c is replaced by $-c$ and their statistic $T(\hat{\rho}-1)$ is identified with U/V; case DK is the moment generating function of White (1958) and Phillips (1987). We write the result in a form that displays its structure.

Proposition 1 (The joint moment generating function, zero start)**.** *Let $\lambda=-s$, $z=2t-c^2$, and let $C=\cos\sqrt{z}$, $S=\sin\sqrt{z}/\sqrt{z}$ and $T=\{S+(C-1)/z\}/z^3$, all even in $\sqrt{z}$ and hence entire functions of z. Then, on the tube,*

$$M(s,t)=e^{-s/2}\,D(s,t)^{-1/2}, \qquad D(s,t)=e^{c}\{F_0(z)+F_1(z)\lambda+F_2(z)\lambda^2\}, \tag{5}$$

where, in case DK,

$$F_0 = C - cS, \qquad F_1 = S, \qquad F_2 = 0;$$

in case DU,

$$F_2 = \frac{S}{z} + \frac{2(C-1)}{z^2}, \qquad F_1 = -c^2 F_2,$$
$$F_0 = \frac{(2t+c^3)S}{z} - \frac{c^2 C}{z} + \frac{4ct(C-1)}{z^2};$$

and in case DT,

$$F_2 = -\frac{4(c^2-3c+27)S}{z^2} + \{192t + 288(1-c)\}T - \frac{32(c^2-3c+6)C}{z^3} - \frac{16(c^2-3c-3)}{z^3},$$
$$F_1 = c^4\Big\{\frac{S}{z^2} - 24T + \frac{8C}{z^3} + \frac{4}{z^3}\Big\},$$
$$F_0 = \frac{\{-c^5 - 8(c^2+3c-3)t\}S}{z^2} - 96(1-c)t^2 T + \Big\{\frac{c^4}{z^2} - \frac{16c^3 t}{z^3}\Big\}C + \frac{8c^2(3-c)t}{z^3}.$$

Each F_m is an entire function of z, and the square root in (5) is the branch that is continuous on the tube and positive at real arguments.

At $c = 0$ in case DK, $D = \cosh\mu - (s/\mu)\sinh\mu$ with $\mu = \sqrt{-2t}$, the formula of Phillips (1987). The determinants of cases DU and DT are rearrangements of Theorems 1 and 2 of Nabeya and Sørensen (1994), who write the characteristic function of $xV - U$ for all x as $D_j(2\mathrm{i}\theta)^{-1/2}e^{\mathrm{i}\theta}$ with D_j explicit combinations of $\cosh\mu$, $\sinh\mu/\mu$ and powers of μ, $\mu = \sqrt{c^2 - 2\lambda x}$; since the characteristic function of $xV - U$ for all x is the joint characteristic function, (5) follows with $\lambda = -s$, $\lambda x = t$ and $D = D_j$, their kernels being twice U and V, which does not affect the probability they compute, and the normalization being fixed by the known distribution of $\int_0^1 (W - \bar{W})^2 \mathrm{d}r$. The form of the proposition exhibits the determinants as polynomials in λ; we have checked it against the limit of the finite-sample characteristic function of Bao et al. (2026), obtained by Richardson extrapolation from $n = 2^{20}$ and 2^{21}, to 2×10^{-8} in all three cases, at real and complex arguments and for $c \in \{-10, -2.5, 0, 1.5\}$, which is an independent check of the determinant of Model 4, which Nabeya and Tanaka (1990a) derived by computer algebra; Section 8 verifies it against the known probabilities and moments of the limits and against simulation. The F_m are evaluated by their Taylor

series in z when $|z| < 4$, and otherwise from the closed forms written in $\vartheta = (-z)^{1/2}$ and $e^{-2\vartheta}$ with the factor e^{ϑ} taken out, so that nothing overflows.

Two properties of D matter. It is an entire function of z, so that the only branch to choose is that of $D^{-1/2}$, which is fixed by continuity from a real point, where $D > 0$, and it has no zeros on the tube. And it is a polynomial in s of degree one (DK) or two (DU, DT), with coefficients that depend on t alone, which is (4) in another guise: under the tilt e^{tV} the Gaussian functionals of (4) remain Gaussian, with complex covariances, and the moment generating function of a quadratic form of rank r in a Gaussian vector is the inverse square root of a polynomial of degree r in its argument. The following lemma makes this precise and gives the configuration of the roots along the contours of Section 4. Let $t_* = \sup\{t : \mathbb{E}e^{tV} < \infty\}$, the first zero of $D(0,\cdot)$ on the positive real axis, and let $(s_2(\tau), s_1(\tau))$, with $s_2(\tau) = -\infty$ in case DK, be the interval of convergence of $M(\cdot,\tau)$.

Lemma 1 (The roots)**.** *Let $\tau < t_*$ and $\sigma \in (s_2(\tau), s_1(\tau))$. For every t with $\operatorname{Re} t = \tau$, $D(\cdot,t)$ is a polynomial in s of exact degree one (DK) or two (DU, DT), with roots $s_1(t), s_2(t)$ continuous in t; at $t = \tau$ the roots are real, with $s_2(\tau) < 0 < s_1(\tau)$ in cases DU and DT and $s_1(\tau) > 0$ in case DK, and $D(\cdot,\tau) > 0$ between them; and $\operatorname{Re} s_2(t) < \sigma < \operatorname{Re} s_1(t)$, respectively $\sigma < \operatorname{Re} s_1(t)$, for every t on the line. Consequently*

$$M(s,t) = \begin{cases} e^{-s/2}P(t)\{(s_1 - s)(s - s_2)\}^{-1/2} & (\mathsf{DU}, \mathsf{DT}), \\ e^{-s/2}P(t)(s_1 - s)^{-1/2} & (\mathsf{DK}), \end{cases} \tag{6}$$

with $P(t) \neq 0$ continuous on the line and $\operatorname{Re}(s_1 - s_2) > 0$.

In terms of the weights $\kappa_k = 1/(2s_k)$, (6) says that, under the tilt e^{tV}, $R = U + \frac{1}{2}$ is distributed as $\kappa_1 Z_1^2 + \kappa_2 Z_2^2$ with Z_1, Z_2 independent standard normal, $\kappa_2 = 0$ in case DK; the sign conditions of the lemma are $\operatorname{Re}\kappa_1 > 0 > \operatorname{Re}\kappa_2$, respectively $\operatorname{Re}\kappa_1 > 0$. With $\lambda = -s$ the roots are those of $F_0 + F_1\lambda + F_2\lambda^2$, so that s_1, s_2 and $P(t) = \{-e^c F_2(z)\}^{-1/2}$ (DU, DT; $e^c F_2 < 0$ at real points, since D is positive between its roots), $P(t) = \{e^c F_1(z)\}^{-1/2}$ (DK), are explicit in the functions of Proposition 1, and the moment generating function of V is $M_V(t) = M(0,t) = P(t)(-s_1 s_2)^{-1/2}$, respectively $P(t)s_1^{-1/2}$.

3.1. The initial observation

With a fixed start γ_0 the Gaussian functionals of (4) acquire means proportional to γ_0, and R becomes a noncentral quadratic form of the same rank; the logarithm of the joint moment generating function gains a term quadratic in γ_0, which is in closed form. In cases DU and DT the fixed start enters the Fredholm determinants of Nabeya and Sørensen (1994, Theorems 1 and 2) through $f = c\gamma_0$ and $f = c^2\gamma_0$, and in case DK it follows from the Riccati equations of the tilted process (Lemma 2 in Appendix A).

Proposition 2 (The joint moment generating function, fixed start)**.** *Let M_0 and D be the moment generating function and the determinant of the zero start, and let M_{γ_0} be the moment generating function of (U, V) with the start γ_0. Then, on the tube of M_0 and wherever the right side is finite,*

$$\log M_{\gamma_0}(s,t) = \log M_0(s,t) + \gamma_0^2\, Q(s,t), \tag{7}$$

where, in cases DU *($k = 1$) and* DT *($k = 2$),*

$$\begin{gathered} Q(s,t) = c^{2k}e^c\left(t + \tfrac{1}{2}s^2 + cs\right)\frac{g(z)}{D(s,t)}, \\ g = -\frac{S}{z} - \frac{2(C-1)}{z^2} \quad (\mathsf{DU}), \qquad g = \frac{12C}{z^3} - \frac{(24+4z)(C-1)}{z^4} - \frac{(24-z)S}{z^3} \quad (\mathsf{DT}), \end{gathered} \tag{8}$$

both entire in z, and in case DK*, with F_0, F_1 those of the case,*

$$Q(s,t) = \frac{tF_1}{F_0} - \frac{s}{2} + \frac{s}{2F_0^2}\,\frac{1}{1 - sF_1/F_0} \qquad (\mathsf{DK}). \tag{9}$$

As a function of s, Q is rational with simple poles at the roots s_1, s_2 of $D(\cdot, t)$ (at $s_1 = F_0/F_1$ in case DK*), and*

$$\begin{gathered} Q(s,t) - Q(0,t) = \frac{s}{2}\left\{\frac{\mu_1}{1-2s\kappa_1} - \frac{\mu_2}{1-2s\kappa_2}\right\}, \\ \mu_k = -\frac{2c^{2k}\left(t + \frac{1}{2}s_k^2 + cs_k\right)g(z)}{F_2(z)\,(s_1 - s_2)\,s_k^2} \qquad (\mathsf{DU}, \mathsf{DT};\ k = 1, 2), \end{gathered} \tag{10}$$

and in case DK *$Q(s,t) - Q(0,t) = -\frac{s}{2} + \frac{s}{2}F_0^{-2}(1-2s\kappa_1)^{-1}$, the constant $-\frac{s}{2}$ being the term $-\frac{1}{2}\gamma_0^2$ of $R = \frac{1}{2}\{J(1)^2 - \gamma_0^2\}$. Under the tilt e^{tV}, R is distributed as $\frac{1}{2}(X_1^2 - X_2^2)$ with X_1, X_2 independent Gaussian of variances $2\kappa_1$, $-2\kappa_2$ and means m_k with $m_k^2 = \gamma_0^2\mu_k$; in case* DK*, $R = \frac{1}{2}\{J(1)^2 - \gamma_0^2\}$ with $J(1)$ of mean γ_0/F_0 and variance F_1/F_0.*

The partial fractions (10) are the exponent of the moment generating function of a noncentral quadratic form of rank two: under the tilt,

$$\mathbb{E}\big(e^{sR}\big) = \prod_{k=1}^{2}(1-2s\kappa_k)^{-1/2}\exp\Big\{\frac{s}{2}\sum_{k=1}^{2}(-1)^{k-1}\frac{m_k^2}{1-2s\kappa_k}\Big\},$$

and (10) identifies the squared means, the only way in which the means enter; at real t the μ_k are nonnegative, which Section 8 verifies numerically. The weights κ_k do not depend on the start. The term $\gamma_0^2 Q(0,t)$, the correction to $\log M_V$, is $c^{2k}\gamma_0^2 t g(z)/F_0(z)$ in cases DU and DT and $\gamma_0^2 t F_1/F_0$ in case DK. The parameter f of Nabeya and Sørensen (1994) collects the initial value and the local drift or trend, $f = |c\gamma_0 + d_1|$ and $|c^2\gamma_0 + cd_1 + d_2|$: the initial value and the drift enter the demeaned or detrended process only through $\gamma_0(e^{cr}-1)$ and $d_1(e^{cr}-1)/c$, and the trend through $d_2(e^{cr}-1-cr)/c^2$, so that (7) also covers a local drift and trend in the data-generating process, with f^2 in place of $c^{2k}\gamma_0^2$.

The stationary start. With a stationary start the joint moment generating function is the average over $\gamma_0 \sim N(0,\sigma_0^2)$, $\sigma_0^2 = -1/(2c)$, of (7). Since $\gamma_0^2 \sim \sigma_0^2\chi_1^2$, averaging $e^{\gamma_0^2 Q}$ multiplies M_0 by $(1-2\sigma_0^2 Q)^{-1/2} = (1+Q/c)^{-1/2}$, which amounts to replacing D by

$$D_{\mathrm{S}}(s,t) = D(s,t) + c^{2k-1}e^c\left(t + \tfrac{1}{2}s^2 + cs\right)g(z) \qquad (\mathsf{SU},\mathsf{ST}), \tag{11}$$

again a polynomial of degree two in s, and in case SK by

$$D_{\mathrm{S}}(s,t) = e^c\Big\{F_0(z) + \frac{S(z)}{2c}\,(2t+s^2)\Big\} \qquad (\mathsf{SK}), \tag{12}$$

of degree two in s with roots $\pm s_2$ and no linear term, as the reversibility of the stationary process requires: $(J(0), J(1))$ is exchangeable, and so is $J(1)^2 - J(0)^2$ with its negative, also under the tilt by $\int_0^1 J^2 \mathrm{d}r$. Lemma 1 holds for D_{S} in all three stationary cases with the configuration of cases DU and DT, and the factorization (6) with $P(t)$ the inverse square root of the leading coefficient; t_* is the first zero of $D_{\mathrm{S}}(0,\cdot)$. Everything that follows applies to the six cases with the appropriate D, and to the fixed start with M_{γ_0} in place of M, where φ_t is then the density of the noncentral form of Proposition 2 under the tilt, given in Section 4.

Closed forms free of cancellation. For $c>0$ the terms of $D(s,t)$ are of the size e^{2c} while the value is of the order of one, and (5) loses about $2c/\ln 10$ digits; at $c=5$ this costs 10^{-12}, at $c=10$ it costs 10^{-7}, and at $c=20$ the determinant is lost altogether. Girsanov's theorem gives forms without this cancellation, which we use for $c>2$ and which also show that the integrands of Section 5 are analytic beyond t_*. Under the law of J with a zero start, a functional Φ of the path has $\mathbb{E}_c\Phi(J)=\mathbb{E}\{\Phi(W)\exp[\frac{c}{2}\{W(1)^2-1\}-\frac{c^2}{2}\int_0^1 W(r)^2\mathrm{d}r]\}$. Let $\boldsymbol{X}=W(1)$ in case DK, $\boldsymbol{X}=(W(1),\int_0^1 W\mathrm{d}r)'$ in case DU and $\boldsymbol{X}=(W(1),\int_0^1 W\mathrm{d}r,\int_0^1 rW\mathrm{d}r)'$ in case DT. Then $V=\int_0^1 W(r)^2\mathrm{d}r-\boldsymbol{X}'\boldsymbol{\Gamma}\boldsymbol{X}$ with $\boldsymbol{\Gamma}=0$, $\mathrm{diag}(0,1)$ and $\mathrm{diag}(0,\boldsymbol{G}^{-1})$, $\boldsymbol{G}^{-1}$ the inverse of the Gram matrix of $(1,r)$ on $[0,1]$, with entries $4,-6,-6,12$, and by (4) $R=\frac{1}{2}\boldsymbol{X}'\boldsymbol{K}\boldsymbol{X}$ with $\boldsymbol{K}=1$,

$$\boldsymbol{K}=\begin{pmatrix}1&-1\\-1&0\end{pmatrix}\quad(\mathsf{DU}),\qquad \boldsymbol{K}=\begin{pmatrix}1&2&-6\\2&-12&12\\-6&12&0\end{pmatrix}\quad(\mathsf{DT}).$$

With $\vartheta=(c^2-2t)^{1/2}$ and $\boldsymbol{Q}=\mathrm{diag}(c,0,\dots,0)-2t\boldsymbol{\Gamma}$, for functions h for which the right side converges,

$$\mathbb{E}\big\{e^{tV}h(R)\big\}=e^{-c/2}\,\mathbb{E}\Big\{\exp\Big(-\frac{\vartheta^2}{2}\int_0^1 W(r)^2\mathrm{d}r\Big)\exp\Big(\frac{1}{2}\boldsymbol{X}'\boldsymbol{Q}\boldsymbol{X}\Big)h\Big(\frac{1}{2}\boldsymbol{X}'\boldsymbol{K}\boldsymbol{X}\Big)\Big\},\qquad(13)$$

and under the probability measure with density $(\cosh\vartheta)^{1/2}\exp(-\frac{\vartheta^2}{2}\int_0^1 W^2\mathrm{d}r)$ the vector $\boldsymbol{X}$ is Gaussian with mean zero and covariance matrix $\boldsymbol{\Sigma}$,

$$\Sigma_{11}=\frac{\tanh\vartheta}{\vartheta},\qquad \Sigma_{12}=\frac{1-\operatorname{sech}\vartheta}{\vartheta^2},\qquad \Sigma_{13}=\Sigma_{22}=\frac{\vartheta-\tanh\vartheta}{\vartheta^3},$$
$$\Sigma_{23}=\frac{\vartheta^2-2+2\operatorname{sech}\vartheta}{2\vartheta^4},\qquad \Sigma_{33}=\frac{\vartheta^3/3-\vartheta+\tanh\vartheta}{\vartheta^5},$$

even in ϑ, with the values 1, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{3}$, $\frac{5}{24}$ and $\frac{2}{15}$ of Brownian motion at $\vartheta=0$. It follows that $M_V(t)=e^{-c/2}(\cosh\vartheta\,|\boldsymbol{I}-\boldsymbol{Q}\boldsymbol{\Sigma}|)^{-1/2}$ and that, under the tilt, R is a quadratic form in a Gaussian vector with covariance matrix $\boldsymbol{P}^{-1}$, $\boldsymbol{P}=\boldsymbol{\Sigma}^{-1}-\boldsymbol{Q}$. In case DK this gives

$$M_V(t)=e^{-c/2}\Big(\frac{\vartheta}{\sinh\vartheta}\Big)^{1/2}\eta^{-1/2},\qquad s_1=\eta=\vartheta\coth\vartheta-c,\qquad(14)$$

and in cases DU and DT the weights $2\kappa_1$, $2\kappa_2$ are the nonzero eigenvalues of $\boldsymbol{K}\boldsymbol{P}^{-1}$, that is the roots of $\alpha_2\nu^2-\alpha_1\nu+\alpha_0=0$ with $\alpha_2=|\boldsymbol{P}|$, $\alpha_1=\mathrm{tr}\{\boldsymbol{K}\,\mathrm{adj}(\boldsymbol{P})\}$ and $\alpha_0=-\mathrm{tr}\{\mathrm{adj}(\boldsymbol{K})\boldsymbol{P}\}$

($\alpha_0 = 1$ in case DU), so that

$$s_1 + s_2 = -\frac{\alpha_1}{\alpha_0}, \qquad s_1 - s_2 = \frac{(\alpha_1^2 + 4\alpha_0\alpha_2)^{1/2}}{\alpha_0}, \qquad \frac{M_V(t)}{(-s_1 s_2)^{1/2}} = P(t) = \frac{e^{-c/2}}{\{\alpha_0 \cosh\vartheta\,|\boldsymbol{\Sigma}|\}^{1/2}}. \tag{15}$$

All quantities are computed without cancellation for large c: the entries of $\boldsymbol{\Sigma}$ from $e^{-2\vartheta}$, the difference $\vartheta - c$ as $-2t/(\vartheta + c)$, and $\eta = (\vartheta - c) + 2\vartheta e^{-2\vartheta}/(1 - e^{-2\vartheta})$. The point t_*, where $|\boldsymbol{I} - \boldsymbol{Q}\boldsymbol{\Sigma}|$ vanishes, enters $P(t)$ and the product $M_V(t)\varphi_t$ of Section 4 only through α_2, polynomially, so these are analytic beyond t_*, up to the first zero of $\cosh\vartheta\,|\boldsymbol{\Sigma}|$, α_0 or $\alpha_1^2 + 4\alpha_0\alpha_2$ on the real axis, which is at most $(c^2 + \pi^2/4)/2$.

4. Inversion formulas and the single integral

The distribution of a ratio $X/\sqrt{Y}$ is a double inversion of the joint moment generating function of (X, Y) with an explicit kernel. Let (X, Y) be a random vector with $Y > 0$ almost surely and moment generating function $M(s, t)$, let $Q_0 = (X - c_0 Y)/\sqrt{Y}$ for a real constant c_0, and for $b \in \mathbb{C}$ and $\operatorname{Re} t > 0$ let

$$J_m(b, t) = \int_0^\infty v^m e^{-bv - tv^2}\,\mathrm{d}v, \tag{16}$$

so that, with the principal branch of $\sqrt{t}$,

$$J_0 = \frac{1}{2}\sqrt{\frac{\pi}{t}}\, e^{b^2/(4t)} \operatorname{erfc}\Big(\frac{b}{2\sqrt{t}}\Big), \quad J_1 = \frac{1 - bJ_0}{2t}, \quad J_2 = \frac{J_0 - bJ_1}{2t}. \tag{17}$$

Theorem 1 (Inversion formulas for $(X - c_0 Y)/\sqrt{Y}$)**.** *Let M be finite in a neighborhood of (σ, τ) with $\tau + c_0\sigma > 0$, let I be an open interval, and suppose that*

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |M(\sigma + \mathrm{i}u, \tau + \mathrm{i}w)|\, e^{-\delta u^2}\mathrm{d}u\mathrm{d}w < \infty \quad \textit{for every } \delta > 0, \tag{18}$$

and that for every compact $K \subset I$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |M(\sigma + \mathrm{i}u, \tau + \mathrm{i}w)| \sup_{q \in K} |J_2(q(\sigma + \mathrm{i}u), \tau + c_0\sigma + \mathrm{i}(w + c_0 u))|\mathrm{d}u\mathrm{d}w < \infty. \tag{19}$$

Then Q_0 has a continuous density on I, given for $q \in I$ by

$$f_{Q_0}(q) = \frac{1}{(2\pi\mathrm{i})^2}\int_{\sigma - \mathrm{i}\infty}^{\sigma + \mathrm{i}\infty}\int_{\tau - \mathrm{i}\infty}^{\tau + \mathrm{i}\infty} M(s, t)\, 2J_2(qs, t + c_0 s)\mathrm{d}t\mathrm{d}s. \tag{20}$$

If moreover $\sigma > 0$ *and (19) holds with* $J_1(qs, t + c_0 s)/s$ *in place of* $J_2(qs, t + c_0 s)$*, then for* $q \in I$

$$\mathbb{P}(Q_0 > q) = \frac{1}{(2\pi \mathrm{i})^2} \int_{\sigma - \mathrm{i}\infty}^{\sigma + \mathrm{i}\infty} \int_{\tau - \mathrm{i}\infty}^{\tau + \mathrm{i}\infty} M(s,t)\, \frac{2J_1(qs, t + c_0 s)}{s} \mathrm{d}t \mathrm{d}s, \tag{21}$$

and if $\sigma < 0$ *the same integral equals* $-\mathbb{P}(Q_0 \le q)$.

The kernels arise from the change of variables $(x, y) = (pv + c_0 v^2, v^2)$, under which $\{(X - c_0 Y)/\sqrt{Y} \le q\}$ becomes $\{p \le q\}$ and the Jacobian is $2v^2$: the inner integral over v of the inversion formula for the density of (X, Y) is (16) with $b = qs$ and $t + c_0 s$ in place of t. The conditions are mild versions of the absolute integrability of M on the contour, and they are needed: for the limits of Section 2 the density of U is unbounded (at $U = -\frac{1}{2}$ in case DK, where $U = \{W(1)^2 - 1\}/2$), so that $|M|$ decays only like $|\operatorname{Im} s|^{-1/2}$ (DK) or $|\operatorname{Im} s|^{-1}$ (DU, DT) along the contour; in t, $|M|$ decays like $\exp(-\frac{1}{2} \operatorname{Re} \mu)$ with $\operatorname{Re} \mu \sim |\operatorname{Im} t|^{1/2}$, and both conditions hold, with $I = \mathbb{R}$. The proof is in Appendix A. The interval I need not be the whole line: where the density has an isolated singularity, the theorem is applied on the open intervals on either side of it, and the distribution function is continuous across the point as soon as the singularity is integrable; this is how Theorem 2 treats its one exceptional point.

The double integrals (20) and (21) can be evaluated by quadrature, but not efficiently: the kernel $J_2(qs, t)$ is in its Gaussian regime where $\operatorname{Im} s$ and $\operatorname{Im} t$ have opposite signs and $|\operatorname{Im} s|$ is up to a few times $|\operatorname{Im} t|$, and there it oscillates with the phase $q^2 (\operatorname{Im} s)^2/(4 \operatorname{Im} t)$, which no grid resolves in the far field. The polynomial structure of Lemma 1 removes the s-integral altogether. The inverse two-sided Laplace transform of (6) in s at the point x is explicit,

$$\frac{1}{2\pi \mathrm{i}} \int_{\sigma - \mathrm{i}\infty}^{\sigma + \mathrm{i}\infty} M(s,t) e^{-sx} \mathrm{d}s = P(t)\, \varphi_t(X), \qquad X = x + \tfrac{1}{2}, \tag{22}$$

with

$$\varphi_t(X) = \begin{cases} \dfrac{e^{-s_1 X}}{(\pi X)^{1/2}}\, \mathbf{1}\{X > 0\} & (\mathsf{DK}), \\[2ex] \dfrac{1}{\pi}\, e^{-(s_1 + s_2)X/2} K_0\Big(\dfrac{(s_1 - s_2)|X|}{2}\Big) & (\mathsf{DU}, \mathsf{DT}), \end{cases} \tag{23}$$

the first a scaled χ^2_1 density, the second the density of a difference of two scaled χ^2_1 variables, both continued analytically in s_1, s_2 and t; the convergence of the integrals behind (22) requires $\operatorname{Re} s_2 < \sigma < \operatorname{Re} s_1$ and $\operatorname{Re}(s_1 - s_2) > 0$, which Lemma 1 provides. Up to the

normalization, φ_t is the density of R under the tilt e^{tV}: its total mass is $(-s_1 s_2)^{-1/2} = M_V(t)/P(t)$, respectively $s_1^{-1/2}$. Inserting (22) into (20) and (21), with the v-integral of (16) written out, gives the density and the distribution function as single integrals over t. Let

$$X(v) = c_0 v^2 + xv + \tfrac{1}{2}, \tag{24}$$

the argument of φ_t along the parabola $u = xv + c_0 v^2$ in the (u, v)-plane, on whose boundary $U = x\sqrt{V} + c_0 V$.

Theorem 2 (The density and distribution function)**.** *Let $0 < \tau < t_*$. In each of the six cases, with D as in Section 3, and with a fixed start, with φ_t the density (32) or (33) of the noncentral form, the limit τ has a continuous distribution function $\mathcal{F}(x)$ and a density $f(x)$, given by (25) at every x at which the integral converges. It converges for every x, and f is continuous, except in case* DK *with $c_0 > 0$ at $x = -\sqrt{2c_0}$, where it diverges: there the density has a logarithmic singularity, and it is continuous on either side. The formulas are*

$$f(x) = \frac{1}{2\pi \mathrm{i}} \int_{\tau - \mathrm{i}\infty}^{\tau + \mathrm{i}\infty} P(t)\, \Phi_x^f(t) \mathrm{d}t, \qquad \mathcal{F}(x) = \frac{1}{2\pi \mathrm{i}} \int_{\tau - \mathrm{i}\infty}^{\tau + \mathrm{i}\infty} P(t)\, \Phi_x(t)\, \frac{\mathrm{d}t}{t}, \tag{25}$$

where

$$\begin{aligned} \Phi_x^f(t) &= \int_0^\infty 2v^2 e^{-tv^2} \varphi_t\big(X(v)\big) \mathrm{d}v, \\ \Phi_x(t) &= \int_0^\infty 2tv\, e^{-tv^2} \Psi_t\big(X(v)\big) \mathrm{d}v = \Psi_t\big(\tfrac{1}{2}\big) + \int_0^\infty e^{-tv^2}\, (x + 2c_0 v)\, \varphi_t\big(X(v)\big) \mathrm{d}v, \end{aligned} \tag{26}$$

and $\Psi_t(X) = \int_{-\infty}^X \varphi_t(u) \mathrm{d}u$.

The second form of Φ_x is an integration by parts in v; the boundary term at $v = 0$ is the mass of φ_t on $(-\infty, \frac{1}{2})$. At $c_0 = 0$ the parabola is the line $u = xv$, and $\Phi_x(t) = \int_{-\infty}^\infty \varphi_t(u + \frac{1}{2})\, tW_x(u,t) \mathrm{d}u$ with $tW_x(u,t) = \int 2tve^{-tv^2} \mathrm{d}v$ over $\{v > 0 : xv \geq u\}$, which is $1 - e^{-tu^2/x^2}$ for $u < 0$ and 0 otherwise when $x < 0$, and $e^{-t \max(u/x, 0)^2}$ when $x > 0$. The convergence of the v-integrals at infinity needs $\mathrm{Re}(t + c_0 s_d) > 0$, s_d the root that governs the decay of φ_t in the direction in which $X(v)$ moves (s_1 for $X \to +\infty$, s_2 for $X \to -\infty$); this holds on the whole line, because $\mathrm{Re}\, s_1 > 0 > \mathrm{Re}\, s_2$ there by Lemma 1, and c_0 and s_d have the same sign whenever $X \to \pm\infty$ requires it. The conditions of Theorem 1 hold with the kernel $J_2(xs, t + c_0 s)$ for σ with $\tau + c_0 \sigma > 0$.

The transforms (26) are special functions of t, $s_1(t)$ and $s_2(t)$. For case DK let

$$\begin{aligned}\mathcal{P}_k(\beta,\gamma) &= \int_0^\infty X^{k-1/2} e^{-\beta X^2-\gamma X}\mathrm{d}X = \Gamma(k+\tfrac{1}{2})(2\beta)^{-(2k+1)/4} e^{\gamma^2/(8\beta)} D_{-k-1/2}\{\gamma(2\beta)^{-1/2}\},\\ \mathcal{Q}_k(\alpha,\beta) &= \int_0^\infty u^k(1+u)^{-1/2} e^{-\alpha u^2-\beta u}\mathrm{d}u,\end{aligned} \tag{27}$$

for $\operatorname{Re}\beta > 0$ and $\operatorname{Re}\alpha > 0$, the first a parabolic cylinder function, which is a Bessel function of order $\frac{1}{4}$ or $\frac{3}{4}$, $D_{-1/2}(x) = (x/2\pi)^{1/2}K_{1/4}(x^2/4)$ and $D_{1/2}(x) = \frac{1}{2}x(x/2\pi)^{1/2}\{K_{1/4}(x^2/4) + K_{3/4}(x^2/4)\}$, with $D_{-3/2}$ and $D_{-5/2}$ from $D_{\nu+1}(x) = xD_\nu(x) - \nu D_{\nu-1}(x)$; the second an incomplete one, with $\mathcal{Q}_0(0,\beta) = e^\beta(\pi/\beta)^{1/2}\operatorname{erfc}(\beta^{1/2})$ and $\mathcal{Q}_k = (-\partial_\beta)^k\mathcal{Q}_0$. For the other cases let

$$\mathcal{G}_k^\pm(a;t) = \int_{\pm(X-1/2)>0} (X-\tfrac{1}{2})^k e^{-a(X-1/2)^2}\varphi_t(X)\mathrm{d}X, \tag{28}$$

the Gaussian transforms of φ_t over the two half-lines, with $\mathcal{G}_0^+(0;t) + \mathcal{G}_0^-(0;t) = (-s_1s_2)^{-1/2}$, the total mass of φ_t.

Corollary 1 (Closed forms at $c_0 = 0$)**.** *Let $c_0 = 0$ and $y = 1/(4x^2)$. In case* DK*,*

$$\Phi_x(t) = \frac{\operatorname{erf}\{(s_1/2)^{1/2}\}}{s_1^{1/2}} + \frac{e^{-s_1/2}}{(2\pi)^{1/2}}\,\mathcal{Q}_0\Big(\frac{t}{4x^2},\frac{s_1}{2}\Big) - \frac{\mathbf{1}\{x<0\}}{\pi^{1/2}}\,e^{-ty}\,\mathcal{P}_0\Big(\frac{t}{x^2}, s_1 - \frac{t}{x^2}\Big), \tag{29}$$

$$\Phi_x^f(t) = \frac{e^{-s_1/2}}{2x^3(2\pi)^{1/2}}\,\mathcal{Q}_2\Big(\frac{t}{4x^2},\frac{s_1}{2}\Big) + \frac{2\,\mathbf{1}\{x<0\}}{|x|^3\pi^{1/2}}\,e^{-ty}\Big(\mathcal{P}_2 - \mathcal{P}_1 + \frac{\mathcal{P}_0}{4}\Big)\Big(\frac{t}{x^2}, s_1 - \frac{t}{x^2}\Big). \tag{30}$$

In cases DU *and* DT*, and in the three stationary cases,*

$$\Phi_x(t) = \begin{cases} (-s_1s_2)^{-1/2} - \mathcal{G}_0^+(0;t) + \mathcal{G}_0^+(t/x^2;t) & (x>0),\\ \mathcal{G}_0^-(0;t) - \mathcal{G}_0^-(t/x^2;t) & (x<0),\end{cases} \qquad \Phi_x^f(t) = \frac{2\,\mathcal{G}_2^\pm(t/x^2;t)}{|x|^3}, \tag{31}$$

with $\mathcal{G}^+$ for $x>0$ and $\mathcal{G}^-$ for $x<0$.

The structure of (29) is that of the general case. The terms without the factor e^{-ty} are analytic in t and vary on the scale of t itself; the terms with it, present for $x<0$ only, are the contributions of a neighborhood of the support edge $X=0$ of φ_t, where the kernel e^{-tu^2/x^2} takes the value e^{-ty}, and they oscillate in $\operatorname{Im}t$ with the period $2\pi/y = 8\pi x^2$. In the other cases the edge is the logarithmic singularity of K_0 at $X=0$, and the same two parts are present inside $\mathcal{G}_0^-(t/x^2;t)$ and $\mathcal{G}_2^-(t/x^2;t)$. With $c_0 \neq 0$ the edge $X=0$ is met at the positive

roots of $X(v)$, of which there are none, one or two: for $c_0 < 0$ exactly one, for every x, so that the oscillating part is present also for $x > 0$; for $c_0 > 0$ two when $x < -\sqrt{2c_0}$, between which $X < 0$ and φ_t vanishes in case DK, and none otherwise. Each root v_j carries the factor $e^{-tv_j^2}$, oscillating in $\operatorname{Im} t$ with the period $2\pi/v_j^2$. In case DK with $c_0 > 0$ the two roots merge at $x = -\sqrt{2c_0}$, where the density has a logarithmic singularity: the t-ratio centered at an explosive null value has an infinite density at the point $-\sqrt{2c_0}$, the largest value that $\tau = -(\frac{1}{2} + c_0 V)/\sqrt{V}$ takes on the event $J(1) = 0$, where $R = 0$. The separation of the two parts, and the evaluation of all the transforms to twelve digits, is the subject of Appendix B.

The initial observation. With a fixed start γ_0, (22) holds with M_{γ_0} in place of M, and φ_t is the density of the noncentral form of Proposition 2 under the tilt, normalized as before. In case DK, $R + \frac{1}{2}\gamma_0^2 = \frac{1}{2}J(1)^2$ with $J(1) \sim N(m, 1/s_1)$, $m = \gamma_0/F_0$, so that, with $\tilde{X} = X + \frac{1}{2}\gamma_0^2$,

$$\varphi_t(X) = e^{\gamma_0^2 t F_1/F_0}\, e^{-s_1 m^2/2}\, \frac{e^{-s_1\tilde{X}} \cosh\{m s_1 (2\tilde{X})^{1/2}\}}{(\pi\tilde{X})^{1/2}}\, \mathbf{1}\{\tilde{X} > 0\}, \tag{32}$$

the support edge having moved to $\tilde{X} = 0$, and $\Psi_t(\frac{1}{2}) = e^{\gamma_0^2 t F_1/F_0} s_1^{-1/2} \bar{F}\{(1+\gamma_0^2)^{1/2}; m, 1/s_1\}$ with $\bar{F}$ as in (41) below. In cases DU and DT, $R = \frac{1}{2}(X_1^2 - X_2^2)$ with $X_1 \sim N(m_1, 1/s_1)$ and $X_2 \sim N(m_2, -1/s_2)$ independent, $m_k^2 = \gamma_0^2 \mu_k$, and integrating out X_2 gives

$$\begin{aligned} \varphi_t(X) &= \frac{2}{\pi}\, e^{\gamma_0^2 Q(0,t)}\, e^{(s_2 m_2^2 - s_1 m_1^2)/2}\, e^{-s_d X}\, G(\pm X; s_1 - s_2, b_1, b_2), \\ G(Y; a, b_1, b_2) &= \int_0^\infty e^{-al^2/2} \cosh(b_1 l)\, \frac{\cosh\{b_2 (2Y + l^2)^{1/2}\}}{(2Y + l^2)^{1/2}} \mathrm{d}l, \end{aligned} \tag{33}$$

with the upper sign, $s_d = s_1$, $b_1 = m_2 s_2$ and $b_2 = m_1 s_1$ for $X > 0$, and the lower sign, $s_d = s_2$, $b_1 = m_1 s_1$ and $b_2 = m_2 s_2$ for $X < 0$; $G(Y; a, 0, 0) = \frac{1}{2} e^{aY/2} K_0(aY/2)$ gives back (23), the sign of the means is immaterial, and G is analytic in Y on $\operatorname{Re} Y > 0$, which is where the paths of Appendix B evaluate it. The kink of φ_t stays at $X = 0$, so the breakpoints and the partition are those of the zero start. What changes is that φ_t is a Gaussian in $(2\tilde{X})^{1/2}$ centered at the mean, so that its mass sits at a distance from the edge, by the factor e^{λ} with $\lambda = m^2 s_d/2$ (12 at $c = -5$ and $\gamma_0 = 1$ in case DU), and the paths are laid out from the exponent $-tv^2 - s_d X(v) + \log\cosh\{b_2 (2\tilde{X}(v))^{1/2}\}$ rather than from its first two terms. The transforms (26) are then evaluated by quadrature as with $c_0 \neq 0$, with G by a 40-point rule, so that the inner integral is a double integral, as it is in the other order.

At $c = c_0 = 0$ in case DK, Theorem 2 can be compared with Abadir (1995). His starting point, the joint density after the inversion in the argument of U (his (2.13)), is (22)–(23) at $c = 0$. He then carries out the inversion in t analytically: for $x < 0$ by a binomial expansion, which gives a convergent triple series of incomplete gamma functions (his (2.11)), and for $x \geq 0$ by changing the order of integration, which leaves a single integral of a parabolic cylinder function (his (2.17) and (2.18)). Evaluated in 30-digit arithmetic, his distribution function and density agree with (25) to 6×10^{-16} at sixteen points of $[-4, 3]$, and his mode of the density, -0.724, with ours, -0.7235. One statement of his does not survive the comparison: his (3.6), repeated from Abadir (1992, p. 316), puts the density of $\hat{\tau}$ at zero at $(\pi e)^{-1/2} = 0.3422$, on the argument that the densities of $R/\sqrt{S}$ and of R/S at zero both equal that of R at zero; for a ratio, $f_{R/\sqrt{S}}(0) = \mathbb{E}(\sqrt{S} \mid R = 0) f_R(0)$, and the value is $f(0) = 0.325567$, which his own (2.18) gives as well at $x = 0.001$.

5. The second representation: the other order of integration

Theorem 2 and the representation of this section are one double integral, taken in its two orders. Let $R = U + \frac{1}{2}$ and $a(v) = \frac{1}{2} + x\sqrt{v} + c_0 v$, so that $\{\tau \leq x\} = \{R \leq a(V)\}$. By (22), under the tilt e^{tV} the measure of R is $P(t)\varphi_t$, so that $P(t)\Psi_t(a) = \mathbb{E}[e^{tV}\mathbf{1}\{R \leq a\}]$ is the Laplace transform in v of the measure $\mathbb{E}[\delta(V - v)\mathbf{1}\{R \leq a\}]$, and for real $\sigma < t_*$

$$\begin{aligned} \mathcal{F}(x) &= \mathbb{P}\{R \leq a(V)\} = \int_0^\infty H(v; x)\mathrm{d}v, \\ H(v; x) &= \mathbb{E}\big[\delta(V - v)\,\mathbf{1}\{R \leq a(v)\}\big] = \frac{1}{2\pi\mathrm{i}} \int_{\sigma - \mathrm{i}\infty}^{\sigma + \mathrm{i}\infty} P(t)\,\Psi_t\big(a(v)\big)\,e^{-tv}\mathrm{d}t. \end{aligned} \tag{34}$$

With the v-integral taken inside, on one line $\mathrm{Re}\, t = \tau$ for every v, the substitution $v = w^2$ gives $\int_0^\infty \Psi_t(a(v))e^{-tv}\mathrm{d}v = \Phi_x(t)/t$, and (34) is the distribution function of Theorem 2; the density is the same with $\sqrt{v}\,\varphi_t(a(v))$ in place of $\Psi_t(a(v))$. This section takes the t-integral inside. Nothing is gained or lost in dimension: $\Phi_x(t)$ and $H(v; x)$ are the two partial integrals of one function of (t, v), and both are evaluated by quadrature. What changes is the contour. Inside the v-integral the line in t is fixed, and the integrand of Φ_x along v is a Gaussian transform of a density that is explicit, or with a fixed start a one-dimensional integral, whose analytic structure is known once and for all. Inside the t-integral the line

may be moved, for each v, to a path through the saddle point of $\log M_V(t) - tv$, on which the integrand decays fast whatever x, at the price of a new path for every v. This is the route of the finite-sample theory in Bao et al. (2026), where it is the natural one, and in the limit it is an independent computation of everything: it shares with Theorem 2 only Proposition 1, in different forms, and nothing of the quadrature. Under the tilt e^{tV}, for real $t < t_*$ and by analytic continuation on the contours, R is $\kappa_1 Z_1^2 + \kappa_2 Z_2^2$ with the weights $\kappa_k = 1/(2s_k)$ of Lemma 1, and its normalized density is $(-s_1 s_2)^{1/2}\varphi_t$, respectively $s_1^{1/2}\varphi_t$; so $M_V(t)$ times that density is $P(t)\varphi_t(r)$, and the joint density of (R, V) is

$$h(r, v) = \frac{1}{2\pi \mathrm{i}} \int_{\sigma - \mathrm{i}\infty}^{\sigma + \mathrm{i}\infty} P(t)\, \varphi_t(r)\, e^{-tv} \mathrm{d}t, \tag{35}$$

for any real $\sigma < t_*$.

Theorem 3 (The double integrals)**.** *Let $\sigma < t_*$ be real. (i) For every $v > 0$,*

$$H(v; x) = \frac{1}{2\pi \mathrm{i}} \int_{\sigma - \mathrm{i}\infty}^{\sigma + \mathrm{i}\infty} M_V(t)\, e^{-tv}\, \mathcal{P}\big(\kappa_1(t), \kappa_2(t); a(v)\big) \mathrm{d}t, \tag{36}$$

where, with $\phi(l; \omega^2) = (2\pi\omega^2)^{-1/2} e^{-l^2/(2\omega^2)}$ and $l_0 = \sqrt{\max(0, -2a)}$,

$$\mathcal{P}(\kappa_1, \kappa_2; a) = 2 \int_{l_0}^{\infty} \phi(l; -2\kappa_2)\ \mathrm{erf}\left(\sqrt{\frac{2a + l^2}{4\kappa_1}}\right) \mathrm{d}l \tag{37}$$

in cases DU *and* DT*, and $\mathcal{P}(\kappa_1, 0; a) = \mathrm{erf}(\sqrt{a/(2\kappa_1)})$ for $a > 0$ and 0 for $a \le 0$ in case* DK*; (37) requires $\mathrm{Re}\,\kappa_1 > 0 > \mathrm{Re}\,\kappa_2$ on the line of integration, which Lemma 1 provides. The line may be replaced by any path Γ that crosses the real axis once, at σ, lies in $\mathrm{Re}\,t \ge \sigma$ and tends to infinity in the lower and the upper half-plane at its two ends, provided the sign conditions hold on Γ and on the region between Γ and the line, and the integrals of the integrand over the parts of the circles $|t| = \varrho$ between Γ and the line tend to zero as $\varrho \to \infty$.*

(ii) With h as in (35) and $a'(v) = x/(2\sqrt{v}) + c_0$, the density and distribution function of the limit are the double integrals

$$f(x) = \int_0^{\infty} \sqrt{v}\ h\big(a(v), v\big) \mathrm{d}v, \tag{38}$$

$$\mathcal{F}(x) = \mathcal{P}\big(\kappa_1(0), \kappa_2(0); \tfrac{1}{2}\big) + \int_0^{\infty} a'(v)\, G\big(a(v), v\big) \mathrm{d}v, \tag{39}$$

$$G(r,v) = \frac{1}{2\pi \mathrm{i}} \int_{\sigma'-\mathrm{i}\infty}^{\sigma'+\mathrm{i}\infty} P(t)\, \varphi_t(r)\, \frac{e^{-tv}}{t} \mathrm{d}t, \tag{40}$$

where $0 < \sigma' < t_*$*; the first term is* $\mathbb{P}(R \le \frac{1}{2})$*, and* $G(r,v) = \int_v^\infty h(r,v')\mathrm{d}v'$*. The lines may be replaced by paths* Γ *as in (i); for* G*, a path that crosses the real axis at a negative point adds the residue* $(-s_1 s_2)^{1/2}\varphi_0(r)$*, the density of* R *without tilt.*

For real weights $\kappa_1 > 0 > \kappa_2$, $\mathcal{P}$ is the probability that $\kappa_1 Z_1^2 + \kappa_2 Z_2^2 \le a$, conditioning on Z_2; for the complex weights on the line of integration it is the analytic continuation of that probability in (κ_1, κ_2), with no branch ambiguity, since the square roots in (37) are of nonnegative reals and of numbers with positive real part. Part (ii) replaces the probability $\mathcal{P}$, itself an integral, by the density of R under the tilt, so that both the density and the distribution function are double integrals with closed-form integrands. The density (38) integrates the joint density of (R, V) along the curve $r = a(v)$, as in the construction of Daniels and Young (1991) for the Studentized mean; the distribution function (39) follows from (34) by an integration by parts in v. Integrating over v first instead, as in (34), gives back Theorem 2, with Φ_x and Φ_x^f as the inner integrals. In case DK the probability $\mathcal{P}$ is an error function, so part (i) is already a double integral; we use it there, because at the support edge $a(v) = 0$ the density φ_t has an inverse square-root singularity and part (ii) is then accurate only to about 10^{-8}. In cases DU and DT we use part (ii). The proof is in Appendix A.

The initial observation. With a fixed start, R is the noncentral form of Proposition 2 under the tilt, and the theorem holds with $M_{\gamma_0}(0,t)$ in place of M_V, the weights unchanged, and $\mathcal{P}$ replaced by

$$\mathcal{P}(\kappa_1, \kappa_2; a) = 2 \int_{l_0}^\infty \bar{\phi}(l; m_2, -2\kappa_2)\, \bar{F}\big((2a + l^2)^{1/2}; m_1, 2\kappa_1\big) \mathrm{d}l, \tag{41}$$

where $\bar{\phi}(l; m, w) = \frac{1}{2}\{\phi(l - m; w) + \phi(l + m; w)\}$ and $\bar{F}(r; m, w) = \frac{1}{2}\{\mathrm{erf}[(r - m)/(2w)^{1/2}] + \mathrm{erf}[(r + m)/(2w)^{1/2}]\}$, the probability that $X^2 \le r^2$ for $X \sim N(m, w)$; in case DK, $\mathcal{P} = \bar{F}((2a + \gamma_0^2)^{1/2}; \gamma_0/F_0, F_1/F_0)$ for $2a + \gamma_0^2 > 0$ and 0 otherwise. Both are even in the means, so only $m_k^2 = \gamma_0^2 \mu_k$ enters, and they are analytic in the weights and the squared means under the sign conditions. Part (ii) holds with the density of R under the tilt in place

of $(-s_1 s_2)^{1/2}\varphi_t$: in case DK with a fixed start it is $\{\phi(\tilde{r}-m;w)+\phi(\tilde{r}+m;w)\}/\tilde{r}$ with $\tilde{r}=(2r+\gamma_0^2)^{1/2}$, $m=\gamma_0/F_0$ and $w=F_1/F_0$, on $2r+\gamma_0^2>0$, and with a stationary start it is (23) with the roots of D_{S}. So five of the six configurations are double integrals with closed-form integrands: case DK with a zero or a fixed start through part (i), and the stationary starts and cases DU and DT with a zero start through part (ii). In cases DU and DT with a fixed start the density of R under the tilt is (33), itself an integral (the density of a product of two correlated noncentral normal variables, which has no closed form: a double series of Whittaker functions, by expanding each noncentral χ_1^2 as a Poisson mixture of central χ_{1+2j}^2); there part (i) is used with (41), a triple integral, with the squared means from (10). With a fixed start the support edge of case DK moves to $a(v)=-\gamma_0^2/2$, where $J(1)=0$, and in the other cases the kink of the density of R stays at $a(v)=0$, where $X_1^2=X_2^2$.

Analyticity beyond t_*. The integrands of the double integrals with a zero start are analytic beyond t_* by the Girsanov forms of Section 3.1. In case DK, for $a>0$,

$$M_V(t)\,\mathcal{P}(\kappa_1,0;a)\,e^{-tv}=e^{-c/2}\Big(\frac{\vartheta}{\sinh\vartheta}\Big)^{1/2}\frac{\operatorname{erf}\{(a\eta)^{1/2}\}}{\eta^{1/2}}\,e^{-tv}, \tag{42}$$

and the factor $\eta^{-1/2}$ of M_V, which is infinite at t_*, where $\eta=0$, cancels: $\operatorname{erf}(w)/w$ is an entire function of w^2, so (42) is analytic up to the first zero of $\sinh\vartheta/\vartheta$, at $t=(c^2+\pi^2)/2$. In cases DU and DT the product $P(t)\varphi_t(r)$ of (15) is analytic up to $(c^2+\pi^2/4)/2$ at most, where $\cosh\vartheta=0$. This matters for $c>0$. There the saddle point of the whole integrand on the real axis often lies beyond t_*, while a path through the saddle point of $\log M_V(t)-tv$ alone must cross the axis to the left of t_*; along such a path the factor $\mathcal{P}$ or $\varphi_t(a(v))$ varies over many orders of magnitude, and the integral loses accuracy (in case DK with $c_0=0$ the error of that route at $x=0$ is 2×10^{-9} at $c=2$ and 5×10^{-3} at $c=5$, where the probability is 0.0170; in cases DU and DT it fails at $c=5$). With a zero start we therefore integrate, for each v, along a path through the saddle point of the logarithm of the whole integrand. With an initial observation the integrands are not of this form, and we integrate along a parabola through the saddle point of $\log M_V(t)-tv$, with the sign conditions checked at every node; this route is accurate to 10^{-11} for $c\le2$, and the programs warn when their error estimates exceed 10^{-10}, as they do by $c=3$ with a fixed start in cases DU and DT. The single integral

of Theorem 2 has no such limitation, with a fixed start as well, because its contour is the fixed line $\operatorname{Re} t = \tau < t_*$ on which the factorization of Lemma 1 holds, and it is the route we take, except far in the tails, where its error, absolute rather than relative, shows, and the double integral takes over (Section 7).

6. Moments

The moments of the limits need no inversion. The gamma integral

$$V^{-p/2} = \frac{1}{\Gamma(p/2)} \int_0^\infty s^{p/2-1} e^{-sV} \mathrm{d}s$$

turns them into one-dimensional integrals of the moment generating function at real arguments.

Proposition 3 (Moments of the limit)**.** *Let $\tau = (U - c_0 V)/\sqrt{V}$ and let $p \geq 1$ be an integer. Then*

$$\mathbb{E}(\tau^p) = \frac{1}{\Gamma(p/2)} \int_0^\infty s^{p/2-1}\, m_p(s) \mathrm{d}s, \qquad m_p(s) = \frac{\partial^p M(u, -s - c_0 u)}{\partial u^p}\Big|_{u=0}, \tag{43}$$

with M the moment generating function of the case: (5) with a zero start, (7) with a fixed start, and (5) with D_{S} in place of D with a stationary start.

The moments exist for every p: by the Cauchy–Schwarz inequality

$$\mathbb{E}\big(|U - c_0 V|^p e^{-sV}\big) \leq \big\{\mathbb{E}(U - c_0 V)^{2p}\, M(0, -2s)\big\}^{1/2},$$

and $M(0, -2s) = D(0, -4s - c^2)^{-1/2}$ decays like $e^{-s^{1/2}}$ up to a power of s, so the integral of $s^{p/2-1}\mathbb{E}(|U - c_0 V|^p e^{-sV})$ converges at both ends; Fubini's theorem gives (43) from the gamma integral, and differentiation under the expectation, which the moment generating function permits near $u = 0$, gives the inner derivative. The moment generating function is analytic in u in the disc $|u| < \min_k |s_k(-s)|$ around 0, by Lemma 1 applied on the real line $t = -s$, and the derivatives are computed as Cauchy integrals on a circle of 0.4 times that radius with 64 nodes (smaller when $c_0 \neq 0$, so that the argument $-s - c_0 u$ of V stays in the domain, and with an initial observation), the branch of $D^{-1/2}$ continued around the

circle from its real point; for $c > 0$ the Girsanov forms of Section 3.1 replace (5), which at $c = 5$ would leave relative errors of order 10^{-6} in the fourth moment. The integral over s is the trapezoidal rule in $\log s$ on $[e^{t_0}, e^8]$, $t_0 = \min(-60, -2c - 40)$, with the step 0.05, whose error, estimated from the rule with twice the step, is below 10^{-10} for the first four moments. A set of four moments takes a fraction of a second.

7. Numerical evaluation

Both representations are evaluated to eleven or twelve digits by the MATLAB programs `ltratio_cdf`, `ltratio_pdf`, `ltratio_icdf` and `ltratio_mom` that accompany the paper, and by the implementations in Julia and in C, the latter with the same MATLAB interface; the single integral is the default, and the double integral is selected by an option (in the MATLAB programs it is the only route with a fixed nonzero start in cases DU and DT, whose noncentral kernel the Julia and C implementations carry, and it is used automatically for the far tails). This section outlines the two computations; Appendix B gives the details.

The single integral. The line of (25) is $\operatorname{Re} t = \tau$ with $\tau = 1$ for $c \le 0$ and a zero start, where t_* exceeds 1 in the three cases (1.23, 4.93 and 19.7 at $c = 0$, increasing as c decreases), and $\tau = \min(1, t_*/2)$ otherwise (t_* is 0.046, 0.10 and 0.97 at $c = 3$, 4×10^{-7} to 10^{-6} at $c = 10$, and 15.1, 23.8 and 37.8 with a stationary start at $c = -5$). The integrand is conjugate symmetric on the line, so the integral is twice the real part of the integral over $\operatorname{Im} t = w > 0$. Its two parts are treated differently. The part of $P(t)\Phi_x(t)/t$ without the oscillating factors is analytic in a strip around the line and decays like $\exp(-\frac{1}{2}\sqrt{w})$, the decay of P: with $w = \tau \sinh u$ the trapezoidal rule in u converges geometrically, and 123 nodes with the step 0.075, up to $w = 4900$, give 10^{-15}. Each part with a factor $e^{-tv_j^2}$, v_j a root of $X(v)$ ($v_j = 1/(2|x|)$ at $c_0 = 0$ and $x < 0$), is written as $G_j(w)e^{-\mathrm{i}wv_j^2}$ with G_j smooth and integrated by Filon's method: G_j is interpolated by Legendre polynomials on panels that grow geometrically from the scale $\min(\frac{1}{2}, 1/v_j^2)$ of the pole at $t = 0$, with 20 Gauss–Legendre nodes each, and the oscillation is integrated exactly, $\int_{-1}^{1} P_n(x)e^{-\mathrm{i}\Omega x}\mathrm{d}x = 2(-\mathrm{i})^n j_n(\Omega)$. The parts are separated inside the transforms (26) by an entire partition of unity, so that each remains analytic in t, and the transforms themselves are evaluated by Gauss–Legendre rules

on paths in the complex v-plane on which the Gaussian factor decays instead of oscillating, with the function $e^z K_0(z)$ evaluated by series and asymptotic expansions to 10^{-14}. For $c > 0$, and with $c_0 < 0$ in the upper tail, the exponent $-tv^2 - s_d X(v)$ of the integrand has a saddle point whose phase grows like x^2 along the line; where its change between the nodes of the trapezoidal rule exceeds $\frac{1}{4}$, the smooth part is integrated instead on geometric panels with Gauss–Legendre rules whose node counts follow the phase. With $c_0 \neq 0$ the roots of $X(v)$, and for $c_0 > 0$ the vertex of the parabola, are breakpoints of the inner integral, and the paths between them are straight pieces along the directions of steepest descent of the exponent at their ends, joined through the real axis where a direct connection would cross a region in which the Gaussian factor grows.

The double integral. For each v the inner inversion runs along a path through a saddle point. With a zero start the integrand is the product (42) or $P(t)\varphi_t(a(v))e^{-tv}$ in the Girsanov forms, analytic beyond t_*; its saddle point $\hat{s}$ on the real axis is found by a safeguarded secant iteration, and the integral is taken along the better, by a nested-rule error estimate, of two paths through $\hat{s}$: a parabola that turns into a wedge, and a vertical line in the ϑ-plane, a parabola in t opening to the right. In case DK a third route writes erf $= 1 -$ erfc and inverts the density of V and the remainder separately, each through its own saddle point. With an initial observation the integrand is M_V times $\mathcal{P}$ of (41), or times the tilted density, and the path is the parabola $t(\theta) = s^* + \mu\theta^2 - 2\mathrm{i}\mu\theta$ of the Talbot method (Weideman and Trefethen, 2007) through the saddle point s^* of $\log M_V(t) - tv$, which opens to the right around the real singularities of M_V, with μ at least 4.5, so that $\mathrm{Re}(-z)^{1/2} \geq 3$ along it, the branch of $\log D$ continued from s^*, and the sign conditions of Theorem 3 checked at every node. Each inner integral uses the trapezoidal rule with 96 nodes after a sinh substitution, with the nested rule of twice the step as error estimate, and the node count is doubled up to twice where the estimate requires it. The outer integral over v uses Gauss–Legendre rules in $\log v$ over $[e^{-8}\mathbb{E}V, e^{6}\mathbb{E}V]$ in parts of length at most two, with breakpoints at the zeros of $a(v)$, where a cosine map removes the singularities of the integrand, graded toward the sharp step of the integrand in the far upper tail for $c > c_0$, and bisected where an error estimate from the Legendre coefficients of the integrand exceeds 10^{-12}; in (39) the boundary term becomes

$\mathbb{P}\{R \le a(v_0)\}$ at the lower end v_0. The programs return an error estimate with each value and warn when it exceeds 10^{-10}.

Cost. Both representations need about 10^5 evaluations of the integrand per value, and their costs differ by the contour alone. In the C implementation, on ten threads of a laptop, a value of the distribution function or the density by the single integral takes 2 to 9 milliseconds at $c_0 = 0$ in all six cases, 5 to 16 milliseconds with $c_0 \neq 0$, rising to 26 in the upper tail at $c = 5$, $c_0 = -5$, where the saddle part needs three outer nodes per radian of its phase, and 13 to 110 milliseconds with a fixed start; the double integral takes 3 to 10 milliseconds with a zero start for $c \le 2$, 20 to 40 at $c = 5$, 17 with a stationary start and 0.23 to 0.26 seconds with a fixed start (Table 1); on one thread both are three to four times slower. The MATLAB programs take 0.03 to 0.09 seconds per value by the single integral at $c_0 = 0$ on one thread, 0.1 to 0.6 seconds with $c_0 \neq 0$ (up to 3 seconds in the upper tail at $c = 5$, $c_0 = -5$), and by the double integral 0.1 to 0.35 seconds with a zero start, 0.4 with a stationary start and 0.1 (case DK) to 2 seconds (cases DU and DT, the triple integral) with a fixed start. The line of the single integral is the same for every x, so that P and the roots s_1, s_2 along it depend on c alone and are shared between values of x; the Julia implementation exploits this and takes 1 to 2 milliseconds per value at $c_0 = 0$. A quantile, by Newton's method on the distribution function with the density as derivative, takes a few values.

Accuracy. Doubling every node count of the single integral changes the distribution function by at most 4×10^{-13} and the density, relative to its value, by 10^{-11}, over $x \in [-4, 3]$ for the three cases at $c = 0$ and $c = -5$; by at most 2×10^{-12} at $c = -10$ and -20 around the center of the distribution; by at most 3×10^{-12} for $c = 1, 2, 3, 5$ over $x \in [-4, 80]$, where the density integrates to one to 10^{-9} and moving the line to $\tau = t_*/4$ changes nothing beyond 3×10^{-12}; and by at most 6×10^{-13} in the stationary cases at $c = -1$, -5 and -20. The trapezoidal rule in w with the step 0.15 applied to the undivided integrand agrees to the same accuracy. Table 1 sets the two representations against each other, both in the C implementation: over the six cases, both sides of the unit root, the local null values of the checks of Bao et al. (2026), the stationary start and the fixed start, the distribution functions agree to 2×10^{-11} at worst, except at one point far in the upper tail at $c = 5$ in case DK, where the double

integral is off by 10^{-8}, and the densities to 3×10^{-10} relative with a zero or stationary start and to 3×10^{-9} with a fixed start, the latter at the points 2.5 standard deviations out, which is the accuracy of the double integral itself. With a fixed start, at 18 points over the three cases, $c \in [-5, 2]$, $c_0 = 0$ and $c_0 \neq 0$, the two agree to 1.7×10^{-13} in the distribution function and to 4×10^{-11} in the density, again the accuracy of the double integral, and to 7×10^{-14} at $c = -20$ and -5 with γ_0 up to 3, where λ reaches 200; moving the line of the single integral between $\tau = 0.5$ and 2.5 changes its values by 5×10^{-15}. Beyond $c = 2$, where the double integral no longer serves, the single integral with a fixed start is unchanged under node doubling to 10^{-16} and under $\tau \to \tau/2$ to 10^{-15} at $c = 3$, 5 and 10, and in case DK it reproduces the closed form of $\mathbb{P}(\tau \leq 0)$ at $c = 5$ to 3×10^{-16}. The two computations share only Proposition 1, in two different forms, and nothing of the quadrature. The error of the single integral is absolute rather than relative: the terms it sums are of order one, and it carries an error of 10^{-14} to 10^{-21} whatever the size of the value, which finer rules do not reduce (2×10^{-14} in the lower tails of cases DT and ST with $c_0 \neq 0$). The programs therefore recompute by the double integral every value of the single integral below 10^{-3}, a density or a distribution function in the lower tail, so that the relative accuracy holds in the tails as well. Far out, below the error of either method, a computed density can be a tiny negative number; the programs return it as zero.

Known values. In case DK with $c_0 = 0$ the limit at $x = 0$ is known: $\mathbb{P}(\tau \leq 0) = \mathbb{P}\{J(1)^2 \leq 1\} = \operatorname{erf}\{(2w)^{-1/2}\}$ with $w = \operatorname{Var} J(1) = (e^{2c} - 1)/(2c)$, which at $c = 0$ is $2\Phi(1) - 1$; with a fixed start, $\mathbb{P}(\tau \leq 0) = \mathbb{P}\{J(1)^2 \leq 1 + \gamma_0^2\}$ with $J(1) \sim N(\gamma_0 e^c, w)$. Both representations reproduce these to 3×10^{-13} for $c \in \{-30, -10, -2.5, 0, 1, 2, 5, 10, 20\}$ with a zero start, where at $c = 20$ the probability is 1.04×10^{-8}, and to 2×10^{-15} with $\gamma_0 = 1.5$ at $c = -5$ and $c = 2$. In case DU they give $\mathbb{P}(\tau_\mu \leq 0) = 0.957565589880$, against the 0.95756559 that Bao and Kan (2026) obtain from the distribution of the coefficient statistic, since $\{\hat{\tau}_\mu \leq 0\} = \{\hat{\phi} \leq 1\}$. When $c_0 \neq 0$ the probability at $x = 0$ is $\mathbb{P}(U - c_0 V \leq 0)$, which has an independent one-dimensional representation, $-(2\pi\mathrm{i})^{-1} \int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} M(\zeta, -c_0\zeta)\zeta^{-1}\mathrm{d}\zeta$ with $\sigma < 0$, that decays fast along the line because of the factor in V; at $(c, c_0) = (0, -5)$, $(2, 1)$, $(-10, -20)$, $(-2.5, -5)$, $(5, 2)$, $(5, -5)$, $(10, 5)$ and $(10, -5)$ the three computations agree to 1.2×10^{-11} in all three

Table 1: The two representations against each other

		$\max\lvert\mathcal{F}_{\text{single}} - \mathcal{F}_{\text{double}}\rvert$			$\max\lvert f_{\text{single}}/f_{\text{double}} - 1\rvert$			time (s)	
c	c_0	DK	DU	DT	DK	DU	DT	single	double
0	0	$1 \cdot 10^{-14}$	$6 \cdot 10^{-13}$	$1 \cdot 10^{-12}$	$1 \cdot 10^{-12}$	$2 \cdot 10^{-12}$	$4 \cdot 10^{-12}$	0.003	0.010
-5	0	$1 \cdot 10^{-14}$	$3 \cdot 10^{-12}$	$3 \cdot 10^{-12}$	$1 \cdot 10^{-12}$	$5 \cdot 10^{-12}$	$2 \cdot 10^{-11}$	0.003	0.009
-20	0	$9 \cdot 10^{-14}$	$2 \cdot 10^{-12}$	$3 \cdot 10^{-12}$	$1 \cdot 10^{-11}$	$2 \cdot 10^{-10}$	$1 \cdot 10^{-10}$	0.003	0.009
2	0	$3 \cdot 10^{-15}$	$9 \cdot 10^{-13}$	$8 \cdot 10^{-13}$	$1 \cdot 10^{-12}$	$2 \cdot 10^{-12}$	$2 \cdot 10^{-12}$	0.002	0.003
5	0	$2 \cdot 10^{-11}$	$5 \cdot 10^{-14}$	$2 \cdot 10^{-11}$	$9 \cdot 10^{-16}$	$5 \cdot 10^{-14}$	$3 \cdot 10^{-10}$	0.005	0.035
0	-5	$7 \cdot 10^{-16}$	$7 \cdot 10^{-13}$	$1 \cdot 10^{-12}$	$6 \cdot 10^{-12}$	$3 \cdot 10^{-12}$	$4 \cdot 10^{-12}$	0.008	0.003
2	1	$6 \cdot 10^{-14}$	$8 \cdot 10^{-13}$	$5 \cdot 10^{-13}$	$7 \cdot 10^{-11}$	$2 \cdot 10^{-12}$	$1 \cdot 10^{-12}$	0.007	0.004
-2.5	-5	$1 \cdot 10^{-15}$	$2 \cdot 10^{-12}$	$1 \cdot 10^{-12}$	$6 \cdot 10^{-13}$	$4 \cdot 10^{-12}$	$3 \cdot 10^{-12}$	0.007	0.003
5	2	$7 \cdot 10^{-13}$	$9 \cdot 10^{-14}$	$3 \cdot 10^{-9}$	$1 \cdot 10^{-10}$	$5 \cdot 10^{-10}$	$1 \cdot 10^{-10}$	0.005	0.021
5	-5	$1 \cdot 10^{-8}$	$2 \cdot 10^{-10}$	$9 \cdot 10^{-14}$	$2 \cdot 10^{-12}$	$2 \cdot 10^{-12}$	$3 \cdot 10^{-12}$	0.026	0.041
-10	-20	$3 \cdot 10^{-15}$	$3 \cdot 10^{-12}$	$1 \cdot 10^{-12}$	$6 \cdot 10^{-12}$	$5 \cdot 10^{-12}$	$2 \cdot 10^{-11}$	0.006	0.004
2	2	$1 \cdot 10^{-14}$	$8 \cdot 10^{-13}$	$5 \cdot 10^{-13}$	$2 \cdot 10^{-8}$	$2 \cdot 10^{-12}$	$2 \cdot 10^{-12}$	0.008	0.004
-2	-2	$9 \cdot 10^{-16}$	$1 \cdot 10^{-12}$	$1 \cdot 10^{-12}$	$1 \cdot 10^{-12}$	$3 \cdot 10^{-12}$	$2 \cdot 10^{-12}$	0.008	0.010
-5	3	$6 \cdot 10^{-16}$	$3 \cdot 10^{-12}$	$2 \cdot 10^{-12}$	$8 \cdot 10^{-13}$	$6 \cdot 10^{-12}$	$3 \cdot 10^{-12}$	0.009	0.009
		SK	SU	ST	SK	SU	ST		
-5	0	$2 \cdot 10^{-12}$	$2 \cdot 10^{-12}$	$2 \cdot 10^{-12}$	$3 \cdot 10^{-12}$	$4 \cdot 10^{-12}$	$7 \cdot 10^{-11}$	0.003	0.017
-20	0	$6 \cdot 10^{-12}$	$7 \cdot 10^{-12}$	$7 \cdot 10^{-12}$	$1 \cdot 10^{-11}$	$4 \cdot 10^{-11}$	$2 \cdot 10^{-10}$	0.003	0.017
		DK	DU	DT	DK	DU	DT		
-5	0, $\gamma_0 = 1$	$8 \cdot 10^{-16}$	$2 \cdot 10^{-15}$	$6 \cdot 10^{-14}$	$3 \cdot 10^{-12}$	$5 \cdot 10^{-12}$	$4 \cdot 10^{-10}$	0.076	0.259
-5	0, $\gamma_0 = 3$	$2 \cdot 10^{-12}$	$5 \cdot 10^{-15}$	$8 \cdot 10^{-15}$	$2 \cdot 10^{-11}$	$6 \cdot 10^{-14}$	$5 \cdot 10^{-12}$	0.087	0.248
2	0, $\gamma_0 = 1$	$2 \cdot 10^{-15}$	$9 \cdot 10^{-16}$	$2 \cdot 10^{-14}$	$2 \cdot 10^{-12}$	$8 \cdot 10^{-10}$	$1 \cdot 10^{-9}$	0.013	0.262
-20	0, $\gamma_0 = 1$	$1 \cdot 10^{-13}$	$2 \cdot 10^{-13}$	$2 \cdot 10^{-14}$	$1 \cdot 10^{-11}$	$5 \cdot 10^{-12}$	$5 \cdot 10^{-12}$	0.109	0.229
-5	-5, $\gamma_0 = 1$	$1 \cdot 10^{-15}$	$2 \cdot 10^{-15}$	$5 \cdot 10^{-14}$	$7 \cdot 10^{-12}$	$1 \cdot 10^{-11}$	$3 \cdot 10^{-9}$	0.081	0.239

Zero start (first block), stationary start (second) and fixed start γ_0 (third). The largest differences between Theorem 2 and Theorem 3 over six or seven points per configuration, at the mean plus -2.5 to 2.5 standard deviations; the relative density differences are over the points with $f > 10^{-8}$. Time: the median per value of the distribution function by the C implementation on ten threads of a laptop. Points at which the double integral reports that its error estimate is not met are left out (the single integral is unchanged under node doubling at all of them): at $c = 5$ in case DK the far upper tail ($x = 126$ and 171 with $c_0 = 0$, $x = 103$ with $c_0 = 2$, $x \geq 163$ with $c_0 = -5$), case SK at $c = -20$ ($x \geq -4.6$), and case DK at $x = -3.8$ with $\gamma_0 = 3$, $c = -5$ and with $\gamma_0 = 1$, $c = -20$. At $c = 5$, $c_0 = -5$ in case DK the double integral is off by 10^{-8} at 1.5 standard deviations above the

cases, and at $(20, 10)$, where the probabilities are about 2×10^{-8}, to 10^{-16}. The stationary-start distribution function equals the fixed-start one averaged over $\gamma_0 \sim N(0, \sigma_0^2)$, an identity that neither computation uses: with 24-point Gauss–Hermite quadrature over γ_0, at $c = -5$ and $x = -2$, the two agree to 8×10^{-13}, 8×10^{-14} and 1.2×10^{-13} in cases SK, SU and ST, and as $\gamma_0 \to 0$ the fixed-start distribution functions tend to those of the zero start, to 4×10^{-13} at $\gamma_0 = 10^{-6}$. The closed form (7) agrees with the Richardson limit of the finite-sample moment generating function of Bao et al. (2026) to 4×10^{-9}, the accuracy of that extrapolation, and its partial fractions (10) fit the limit as a function of s with no room for a further linear or quadratic term; at real arguments the squared means μ_k are positive, and the prediction of (10) at a third value of s from two others holds to 10^{-16}. The densities agree with fifth-order central differences of the distribution functions to 4×10^{-13}. Finally, two million paths of the Ornstein–Uhlenbeck process, by Euler's scheme with 1000 steps, give the limiting functionals (3) directly: at $c = 0$ and $c = -5$ with a zero start, and at $c = -5$ and -10 with a stationary start, the simulated distribution functions at $x = -3, -2, -1, 0$ agree with the exact ones to within 0.003 at all 48 points, the differences being the discretization error of the scheme, which at this number of paths exceeds the sampling error of 3×10^{-4} (for $\hat{\tau}_\tau$ at $c = 0$ and $x = -2$, with 4×10^5 paths, the difference is -0.013, -0.007 and -0.003 with 250, 500 and 1000 steps, and within the sampling error of 0.0008 with 2000 and 4000); with a fixed start $\gamma_0 = 1.5$ and with a stationary start, 2×10^5 simulated regressions at $n = 1000$ and $\phi = 1 - 5/n$ agree with the limits within about one standard error at $x \in \{-3, -2, -1, 0\}$ in all six cases.

8. Numerical results

8.1. Quantiles and critical values

Table 2 gives the quantiles of the three limiting distributions at the unit root and under the local alternative $c = -5$, with a zero start, and the asymptotic critical values of MacKinnon (2010), which agree to within 1.3×10^{-4}, consistent with their standard errors. The Richardson extrapolation of the exact finite-sample quantiles of Bao et al. (2026) from $n = 500$, 1000 and 2000 is within 10^{-5} of these in 22 of the 27 cases at $c = 0$, and off by

Table 2: Quantiles of the limiting Dickey–Fuller t-ratios

	c	1%	2.5%	5%	10%	25%	50%	90%	95%	97.5%	99%
$\hat{\tau}$	0	−2.5658	−2.2274	−1.9409	−1.6167	−1.0913	−0.5001	0.8875	1.2831	1.6230	2.0145
M10		−2.5657		−1.9410	−1.6168						
	−5	−3.2587	−2.9735	−2.7368	−2.4751	−2.0674	−1.6623	−1.0126	−0.8321	−0.6579	−0.4310
$\hat{\tau}_\mu$	0	−3.4304	−3.1221	−2.8616	−2.5668	−2.0874	−1.5664	−0.4403	−0.0785	0.2380	0.6067
M10		−3.4303		−2.8615	−2.5668						
	−5	−3.7216	−3.4345	−3.1943	−2.9258	−2.4999	−2.0632	−1.3233	−1.1126	−0.9143	−0.6634
$\hat{\tau}_\tau$	0	−3.9587	−3.6614	−3.4105	−3.1270	−2.6685	−2.1806	−1.2464	−0.9400	−0.6597	−0.3251
M10		−3.9588		−3.4105	−3.1271						
	−5	−4.1316	−3.8452	−3.6044	−3.3338	−2.9002	−2.4477	−1.6594	−1.4348	−1.2267	−0.9653

Zero start, no drift, $c_0 = 0$. Quantiles of the limits of $\hat{\tau}$, $\hat{\tau}_\mu$ and $\hat{\tau}_\tau$ under $\phi = 1 + c/n$ from Theorem 2; M10: the asymptotic critical values of MacKinnon (2010). To five digits, at the probabilities 0.01, 0.025, 0.05, 0.10, 0.50, 0.90, 0.95, 0.975, 0.99 and $c = 0$: −2.56580, −2.22741, −1.94087, −1.61670, −0.50014, 0.88754, 1.28314, 1.62295, 2.01451 ($\hat{\tau}$), −3.43035, −3.12209, −2.86157, −2.56679, −1.56643, −0.44030, −0.07847, 0.23801, 0.60667 ($\hat{\tau}_\mu$), −3.95868, −3.66140, −3.41045, −3.12702, −2.18060, −1.24636, −0.94004, −0.65970, −0.32511 ($\hat{\tau}_\tau$).

up to 6×10^{-5} in the upper tails, mainly of $\hat{\tau}$, where the finite-sample quantiles approach their limit more slowly. For $\hat{\tau}$ the limiting density has the sharp peak near −0.72 and the shoulder to its right that Abadir (1995) obtained.

8.2. Local alternatives and the local null value

Table 3 sets the limit against the values that Larsson (1995) tabulated for case DK, for the Dickey–Fuller statistic under the explosive local alternative $\phi = e^{\beta/n}$ and for the t-ratio centered at the true value, $c = c_0 = \beta$. His simulated values, from 10^5 series of length 400, agree with the limit to within 0.006 and mostly within their sampling error; the two largest differences, 0.84 against 0.846 at $\beta = 1$, $z = 2$ in panel A and 0.071 against 0.0747 at $\beta = 2$, $z = -2$ in panel B, are of four to five standard errors (in the latter the exact finite-sample probability at $n = 400$ is 0.0756 (Bao et al., 2026), so the finite sample does not account for it). His approximation (3.4) to his exact upper-tail formula is accurate only far in the upper tail: at $\beta = 2$ it gives 0.05 and 0.14 at $z = 0.5$ and 1, where the limit is 0.277 and 0.343.

His saddlepoint tail approximations are close in the lower tail at $\beta = 1$ (0.029 for 0.026 at $z = -2$) but not at $\beta = 2$ (0.003 for 0.015), and in panel B they miss the upper tail (0.59 and 0.92 for 0.961 and 0.987 at $\beta = 2$).

8.3. Explosive alternatives and the right-tailed test

Tests for speculative bubbles use the other tail of the same statistic. Diba and Grossman (1988) and, in the form now standard, Phillips et al. (2011) and Phillips et al. (2015) test the unit root against the explosive alternative $\phi > 1$ by rejecting for large values of $\hat{\tau}_\mu$, over the full sample and over recursive or rolling windows. For the test on a fixed window the null limit is that of Table 2, so its critical values are the upper quantiles there, -0.0785 and 0.6067 at the 5% and 1% levels for $\hat{\tau}_\mu$, which are the values that Phillips et al. (2011) obtain by simulation to two decimals, and its limiting power against the explosive local alternatives $\phi = 1 + c/n$, $c > 0$, is the upper tail of the distributions of Section 4, which cover $c > 0$ as they cover $c < 0$. Table 4 gives it. The power rises steeply between $c = 1$ and $c = 3$, from a fifth to four fifths without deterministic terms, and the test with a trend is blind to the alternative until c exceeds about 1: at $c = 0.5$ and 1 its power is below its size, the explosive drift being absorbed by the trend. The sup statistics of Phillips et al. (2011), which take the maximum over windows, have a different limit, a supremum of the functionals (3) over subintervals, which the present method does not reach.

8.4. Local power and the initial observation

Table 5 gives the limiting power of the 5% tests, with the limiting critical values of Table 2, against $\phi = 1 + c/n$ with a fixed start of ξ stationary standard deviations ($\gamma_0 = \xi/\sqrt{-2c}$) and with a stationary start; for $\hat{\tau}$, whose null distribution itself depends on a start of this order, the critical value is held at its zero-start value. Four features stand out.

- Power rises with the initial deviation, and most without deterministic terms: at $c = -5$ it rises from 0.318 at $\xi = 0$ to 0.720 at $\xi = 3$ for $\hat{\tau}$, from 0.117 to 0.208 for $\hat{\tau}_\mu$, and only from 0.083 to 0.090 for $\hat{\tau}_\tau$. This agrees with the power curves of Müller and Elliott (2003) and the simulations of Harvey et al. (2009), in which the power of the least

Table 3: The limiting distribution of $\hat{\tau}$ under local alternatives: Larsson (1995) and the exact values

A. Dickey–Fuller test, $\mathbb{P}(\hat{\tau} \le z)$ under $\phi = e^{\beta/n}$ ($c = \beta$, $c_0 = 0$; his Table B.3)

	$\beta = 1$				$\beta = 2$			
z	Sim.	(3.4)	Tail	Exact	Sim.	(3.4)	Tail	Exact
−2	0.026		0.029	0.026405	0.015		0.003	0.015333
−1.5	0.076		0.096	0.075778	0.042		0.019	0.043064
−1	0.17		0.30	0.171927	0.093			0.094033
0	0.42			0.424176	0.21			0.215288
0.5	0.55	0.47		0.545650	0.28	0.05		0.277173
1	0.66	0.61		0.661230	0.34	0.14		0.342599
2	0.84	0.83		0.845905	0.48	0.36		0.479265
3	0.948	0.945		0.947926	0.61	0.54		0.611544
5	0.998	0.998		0.997702	0.82	0.80		0.818787
8	1	1		0.999998	0.964	0.960		0.962540

B. t-ratio centered at the true value, $\mathbb{P}(\hat{\tau}_0 \le z)$ with $c = c_0 = \beta$ (his Table B.2)

	$\beta = -2$			$\beta = 2$		
z	Sim.	Tail	Exact	Sim.	Tail	Exact
−2	0.039	0.040	0.039149	0.071	0.081	0.074729
−1.5	0.11	0.10	0.109743	0.19	0.33	0.191272
−1	0.24		0.244738	0.32		0.322495
0	0.65		0.647192	0.65		0.650436
1	0.92		0.917900	0.90		0.901895
1.5	0.97	0.61	0.969410	0.96	0.59	0.960805
2	0.99	0.94	0.990543	0.99	0.92	0.987266
3	0.999	0.998	0.999515	0.999	0.998	0.999303

Case DK, zero start. Sim.: Larsson (1995), Tables B.2 and B.3, column 1, from 10^5 simulated series of length 400 (standard error at most 1.5×10^{-3}); (3.4): his approximation (3.4) to his exact upper-tail formula (3.3); Tail: his saddlepoint tail approximations (Theorem 4.1); blank where he gives none. Exact: Theorem 2.

Table 4: Limiting power of the right-tailed Dickey–Fuller tests against explosive alternatives

	Level	Critical value	c				
			0.5	1	2	3	5
$\hat{\tau}$	5%	1.2831	0.134	0.279	0.619	0.835	0.973
	1%	2.0145	0.047	0.152	0.519	0.792	0.968
$\hat{\tau}_\mu$	5%	−0.0785	0.087	0.170	0.569	0.844	0.978
	1%	0.6067	0.022	0.060	0.423	0.792	0.972
$\hat{\tau}_\tau$	5%	−0.9400	0.048	0.048	0.225	0.707	0.970
	1%	−0.3251	0.010	0.010	0.109	0.616	0.961

$\mathbb{P}(\tau > x)$ under $\phi = 1 + c/n$, zero start, $c_0 = 0$, with x the upper 5% and 1% quantiles of the null limit (Table 2), from Theorem 2.

squares Dickey–Fuller tests grows with the magnitude of the initial deviation, unlike that of the GLS-detrended tests.

- Close to the unit root the effect reverses for $\hat{\tau}_\tau$: at $c = -2.5$ its power falls from 0.0595 at $\xi = 0$ to 0.0570 at $\xi = 3$, and from 0.0595 to 0.0592 with a stationary start.

- The stationary start behaves almost exactly like the fixed start $\xi = 1$, its root mean square: at $c = -5$ the powers are 0.362 and 0.361 for $\hat{\tau}$ and 0.125 and 0.124 for $\hat{\tau}_\mu$.

- The stationary-start powers reproduce the simulations of Elliott (1999). His large-sample powers of the Dickey–Fuller t-tests with a stationary start (his Table 2, from $T = 1000$ and 20,000 replications) are 0.13, 0.33, 0.62 and 0.86 with an intercept and 0.08, 0.19, 0.39 and 0.63 with a trend, at $c = -5$, -10, -15 and -20. The limits are 0.125, 0.325, 0.618 and 0.859, and 0.084, 0.188, 0.379 and 0.621.

The last rows of each panel are the exact powers at $n = 100$ with a stationary start from Bao et al. (2026), which use the exact finite-sample critical values. The limit is an excellent guide to the size-adjusted power at $n = 100$ near the unit root, within 0.005 for $c \geq -5$, and it understates power further away, by up to 0.045 for $\hat{\tau}_\tau$ at $c = -20$; for $\hat{\tau}_\tau$ at $c = -15$ and

-20 the values of Elliott (1999) lie between the limit and the exact power at $n = 100$ (0.404 and 0.664), as a sample of $T = 1000$ would.

8.5. Moments

Table 6 gives the mean and standard deviation of the limit of the Dickey–Fuller statistic under local alternatives, with a zero and with a stationary start, from Proposition 3. At the unit root the first six moments of the three limits agree with the eight-digit values of Nabeya (1999), computed from the formula of Sawa (1978) with Simpson's rule, to within one unit in their last digit; the second moment of $\hat{\tau}_\mu$, 3.05599535, is also the expectation 3.0560 of the limiting likelihood ratio statistic in Larsson (1997). Under local alternatives the first four moments agree with Gauss–Legendre integration of the densities of Theorem 2 to 2×10^{-11} relative ($\hat{\tau}_\mu$ at $c = -5$, $\hat{\tau}_\tau$ at $c = -10$, $\hat{\tau}_\tau$ at $c = c_0 = -5$, and with a stationary start), and on the explosive side to 1.2×10^{-10} at $c = c_0 = 10$; on that side the means of the Dickey–Fuller statistic grow fast, 59.0696374465 and 45.7021278187 in cases DK and DU at $c = 5$. As c decreases the distribution moves to the left and becomes narrower, and the shift is largest without deterministic terms: from $c = 0$ to $c = -5$ the mean falls by 1.28 for $\hat{\tau}$, by 0.56 for $\hat{\tau}_\mu$ and by 0.29 for $\hat{\tau}_\tau$, which is the ordering of the local powers in Table 5. A stationary start moves the mean further to the left for $\hat{\tau}$ at every c, and for $\hat{\tau}_\mu$ and $\hat{\tau}_\tau$ from $c = -5$ and $c = -10$ on; closer to the unit root it moves them slightly to the right, the counterpart of the small loss of power there. It changes the standard deviation by at most 0.016.

9. Conclusion

The limiting distribution of the Dickey–Fuller t-ratio is a single integral of special functions: the Fredholm determinant that generates the numerator and the denominator of the limit is a polynomial of degree at most two in the argument of the numerator, which makes one of the two inversions explicit, and what remains is an integral along a line in the complex plane that is evaluated to twelve digits in a fraction of a second. The representation covers the three regressions, local alternatives on both sides of the unit root, the t-ratio centered at the unit root or at any local null value, and a zero, a fixed or a stationary initial observation; a fixed initial observation enters the moment generating function through a closed-form

Table 5: Limiting power of the 5% Dickey–Fuller tests and the initial observation

	Start	c					
		-30	-20	-15	-10	-5	-2.5
$\hat{\tau}$	$\xi = 0$	1.000	0.998	0.967	0.758	0.318	0.144
	$\xi = 1$	1.000	0.999	0.974	0.792	0.361	0.173
	$\xi = 2$	1.000	0.999	0.988	0.877	0.499	0.277
	$\xi = 3$	1.000	1.000	0.997	0.958	0.720	0.499
	stationary	1.000	0.998	0.973	0.789	0.362	0.176
	stationary, $n = 100$	1.000	0.999	0.976	0.802	0.369	0.178
$\hat{\tau}_\mu$	$\xi = 0$	0.995	0.844	0.593	0.304	0.117	0.072
	$\xi = 1$	0.996	0.860	0.619	0.325	0.124	0.074
	$\xi = 2$	0.998	0.901	0.691	0.391	0.152	0.081
	$\xi = 3$	0.999	0.948	0.795	0.507	0.208	0.098
	stationary	0.996	0.859	0.618	0.325	0.125	0.074
	stationary, $n = 100$	0.997	0.883	0.646	0.337	0.126	0.074
$\hat{\tau}_\tau$	$\xi = 0$	0.940	0.606	0.366	0.182	0.083	0.059
	$\xi = 1$	0.945	0.621	0.379	0.188	0.084	0.059
	$\xi = 2$	0.959	0.666	0.417	0.206	0.086	0.058
	$\xi = 3$	0.975	0.734	0.483	0.241	0.090	0.057
	stationary	0.945	0.621	0.379	0.188	0.084	0.059
	stationary, $n = 100$	0.965	0.664	0.404	0.195	0.084	0.059

$\mathbb{P}(\tau \le x_{0.05})$ under $\phi = 1 + c/n$, no drift, $c_0 = 0$, with $x_{0.05}$ the limiting 5% critical value at the unit root with a zero start (-1.94087, -2.86157, -3.41045). ξ: fixed start $y_0 = \sigma\sqrt{n}\,\xi/\sqrt{-2c}$, from Theorem 2, with (32) and (33) for $\xi > 0$, and checked against Theorem 3 with (41), which agrees to the three decimals shown; stationary: y_0 from the stationary distribution, with D_{S} of Section 3.1; stationary, $n = 100$: the exact finite-sample power of Bao et al. (2026) with the exact critical values.

Table 6: Mean and standard deviation of the limit of the Dickey–Fuller t-ratio under local alternatives

	Start	c				
		0	−2.5	−5	−10	−20
$\hat{\tau}$	zero	−0.4231	−1.2636	−1.7066	−2.3352	−3.2366
		(0.9811)	(0.6548)	(0.5867)	(0.5461)	(0.5239)
	stationary		−1.3590	−1.7775	−2.3874	−3.2747
			(0.6500)	(0.5883)	(0.5490)	(0.5261)
$\hat{\tau}_\mu$	zero	−1.5330	−1.8373	−2.0960	−2.5924	−3.4084
		(0.8403)	(0.6949)	(0.6381)	(0.5836)	(0.5458)
	stationary		−1.8347	−2.1226	−2.6279	−3.4404
			(0.7100)	(0.6424)	(0.5853)	(0.5473)
$\hat{\tau}_\tau$	zero	−2.1814	−2.2927	−2.4738	−2.8644	−3.5905
		(0.7499)	(0.7086)	(0.6640)	(0.6110)	(0.5652)
	stationary		−2.2862	−2.4731	−2.8779	−3.6118
			(0.7133)	(0.6684)	(0.6129)	(0.5664)

Mean and, in parentheses, standard deviation of the limit of the Dickey–Fuller t-ratio ($c_0 = 0$) under $\phi = 1 + c/n$, no drift, from Proposition 3; the stationary start requires $c < 0$. At $c = 0$ the values are those of Nabeya (1999).

quadratic term, and the integrand through the density of a noncentral quadratic form. The same integral in the other order, with the inversion in the argument of the denominator inside, along saddle-point paths, checks the first and serves far in the tails. The moments are one-dimensional integrals. The critical values, local powers and moments that have been obtained by simulation since Dickey and Fuller (1979) are now exact.

The structure exploited here, a quadratic form of low rank in the numerator under a tilt of the denominator, arises wherever a t-ratio is formed from a Gaussian process: the limits of the Dickey–Fuller tests with a drift or a trend in the data-generating process (Nabeya and Sørensen, 1994), which Proposition 2 already covers through the parameter f, and, after

a convolution with a normal, the t-ratio in a predictive regression with a near-integrated regressor; Theorem 1 applies to them without change.

Declaration of the Use of Generative AI

An AI assistant (Claude, Anthropic) was used in drafting and editing the text and in writing and checking the replication code; all results were verified by the authors, who take full responsibility for them.

Appendix A. Proofs

Proof of Theorem 1. Assume first that

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}|M(\sigma+\mathrm{i}u,\tau+\mathrm{i}w)|\mathrm{d}u\mathrm{d}w<\infty. \tag{A.1}$$

Then, by the inversion theorem for the bivariate two-sided Laplace transform, (X,Y) has a continuous density

$$f(x,y)=\frac{1}{(2\pi\mathrm{i})^2}\int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty}\int_{\tau-\mathrm{i}\infty}^{\tau+\mathrm{i}\infty}M(s,t)e^{-sx-ty}\mathrm{d}t\mathrm{d}s,$$

and $|f(x,y)|\le Ce^{-\sigma x-\tau y}$ with C equal to $(2\pi)^{-2}$ times (A.1). The map $(p,v)\mapsto(pv+c_0v^2,v^2)$ takes $\mathbb{R}\times(0,\infty)$ one-to-one onto itself with Jacobian $2v^2$, and $pv+c_0v^2-c_0v^2\le q\sqrt{v^2}$ if and only if $p\le q$, so

$$\mathbb{P}(Q_0\le q)=\int_{-\infty}^{q}\int_0^{\infty}2v^2f(pv+c_0v^2,v^2)\mathrm{d}v\mathrm{d}p.$$

The inner integrand is bounded by $2Cv^2e^{-\sigma pv-(\tau+c_0\sigma)v^2}$, which is integrable locally uniformly in p since $\tau+c_0\sigma>0$; hence the inner integral is continuous in p and is the density of Q_0. Inserting the inversion formula, the absolute integrand is $2v^2|M(s,t)|e^{-\sigma qv-(\tau+c_0\sigma)v^2}$, whose integral is $2J_2(\sigma q,\tau+c_0\sigma)$ times (A.1), so by Fubini the v-integral may be done first; it is $2J_2(qs,t+c_0s)$, which gives (20). For (21), integrate (20) over $p>q$: since $\int_q^\infty\int_0^\infty v^2e^{-\sigma pv-(\tau+c_0\sigma)v^2}\mathrm{d}v\mathrm{d}p<\infty$ for $\sigma>0$, the order of integration may be changed, and

$$\int_q^\infty 2J_2(ps,t+c_0s)\mathrm{d}p=2\int_0^\infty v^2e^{-(t+c_0s)v^2}\int_q^\infty e^{-psv}\mathrm{d}p\mathrm{d}v=\frac{2J_1(qs,t+c_0s)}{s}.$$

For $\sigma < 0$ integrate over $p < q$ instead, where $\int_{-\infty}^{q} e^{-psv}\mathrm{d}p = -e^{-qsv}/(sv)$.

In general, let Z be standard normal and independent of (X, Y), and for $\epsilon > 0$ let $Q_\epsilon = (X + \epsilon Z - c_0 Y)/\sqrt{Y}$. The moment generating function of $(X + \epsilon Z, Y)$ is $M(s,t)e^{\epsilon^2 s^2/2}$, and $|e^{\epsilon^2 s^2/2}| = e^{\epsilon^2(\sigma^2 - u^2)/2}$, so it satisfies (A.1) by (18), and by the first part Q_ϵ has the continuous density $f_\epsilon(q)$ given by (20) with $e^{\epsilon^2 s^2/2}$ inserted. Let $g(q)$ denote the right side of (20). For $\epsilon \le 1$ the integrands of f_ϵ and g are dominated by $e^{\sigma^2/2}$ times the integrand of (19), so g is continuous on I and, by dominated convergence, $f_\epsilon \to g$ uniformly on compact subsets of I as $\epsilon \to 0$. The passage from f_ϵ to the density of Q_0 uses only this locally uniform convergence to a continuous limit, and is stated so that the proof of Theorem 2 can use it: *if the densities f_ϵ of Q_ϵ converge as $\epsilon \to 0$, locally uniformly on I, to a continuous function g, then the distribution function of Q_0 is continuous on I and g is its density there.* Since $Q_\epsilon \to Q_0$ almost surely, hence in distribution, $\mathbb{P}(Q_\epsilon \le q) \to \mathbb{P}(Q_0 \le q)$ at every continuity point q of the distribution function of Q_0 (Billingsley, 1995, Theorem 25.2). So for $a < b$ in I that are continuity points,

$$\mathbb{P}(a < Q_0 \le b) = \lim_{\epsilon \to 0} \mathbb{P}(a < Q_\epsilon \le b) = \lim_{\epsilon \to 0} \int_a^b f_\epsilon(q)\mathrm{d}q = \int_a^b g(q)\mathrm{d}q,$$

the last step by the uniform convergence on $[a, b]$. The discontinuity points are countable, so every $a \in I$ lies between continuity points $a' < a < a''$ as close to a as desired, and $\mathbb{P}(Q_0 = a) \le \int_{a'}^{a''} g(q)\mathrm{d}q \to 0$, g being bounded on compact subsets of I: the distribution function of Q_0 is continuous on I, the identity extends to all $a < b$ in I, and g is the density of Q_0 on I. For (21), the first part gives $\mathbb{P}(Q_\epsilon > q)$ as the integral with $e^{\epsilon^2 s^2/2}$ inserted, which converges to the right side of (21) by dominated convergence, while $\mathbb{P}(Q_\epsilon > q) \to \mathbb{P}(Q_0 > q)$ by the same theorem, q being a continuity point; the case $\sigma < 0$ is the same. □

Proof of Lemma 1. Write the Gaussian path as ξ with covariance operator Σ, $U = \frac{1}{2}\langle \xi, A\xi \rangle - \frac{1}{2}$ and $V = \langle \xi, B\xi \rangle$ with $B \ge 0$ of trace class, and $A = R^* A_1 R$ with R of rank r onto $\mathbb{R}^r$ and A_1 a nonsingular symmetric $r \times r$ matrix: $r = 1$ with $A_1 = 1$ in case DK, where $U = \{J(1)^2 - 1\}/2$, and $r = 2$ with A_1 indefinite in cases DU and DT, where $U + \frac{1}{2}$ is $\frac{1}{2}J(1)^2 - \bar{J}J(1)$ and $\frac{1}{2}J(1)^2 + 2J(1)m_0 - 6J(1)m_1 - 6m_0^2 + 12m_0 m_1$, $m_k = \int_0^1 r^k J\mathrm{d}r$, whose matrices have determinants -1 and 0 with rank 2. By the formula for the moment generating function of

Gaussian quadratic forms, on the tube $M(s,t) = e^{-s/2}\det(I - 2t\Sigma B - 2s\Sigma A)^{-1/2}$, and since $(I - 2t\Sigma B)^{-1}\Sigma = (\Sigma^{-1} - 2tB)^{-1} =: \Sigma_t$ and $\det(I - 2s\Sigma_t R^* A_1 R) = \det(I_r - 2sA_1 R\Sigma_t R^*)$,

$$D(s,t) = \det(I - 2t\Sigma B)\,\det(I_r - 2sA_1 R\Sigma_t R^*),$$

a polynomial in s of degree r with leading coefficient $(-2)^r \det A_1 \det(R\Sigma_t R^*)\det(I - 2t\Sigma B)$. For $\operatorname{Re} t < t_*$ the operator $\Sigma^{-1} - 2tB$ has positive definite real part, hence so has its inverse Σ_t and the $r \times r$ matrix $R\Sigma_t R^*$, which is therefore nonsingular (a complex symmetric matrix G with $v^*(\operatorname{Re} G)v > 0$ for $v \neq 0$ has $v^*Gv \neq 0$); this is the accretivity argument of Bao and Kan (2026). And $\det(I - 2t\Sigma B) = D(0,t) \neq 0$ for $\operatorname{Re} t < t_*$, since $|M_V(t)| \leq M_V(\operatorname{Re} t) < \infty$. So the degree is exactly r and the roots are continuous in t. At real $t = \tau$, Σ_τ is a covariance, $A_1 R\Sigma_\tau R^*$ is similar to the symmetric matrix $(R\Sigma_\tau R^*)^{1/2} A_1 (R\Sigma_\tau R^*)^{1/2}$, whose inertia is that of A_1 by Sylvester's law, and the roots $s_j = 1/(2\lambda_j)$, λ_j its eigenvalues, are real with the signs stated; $D(\cdot,\tau)$ is positive at $s = 0$, vanishes only at the roots, and the component of $\{D > 0\}$ containing 0 is the interval of convergence of $M(\cdot,\tau)$. Finally, $|M(s,t)| \leq M(\sigma,\tau) < \infty$ on the tube, so $D(s,t) \neq 0$ there: no root crosses the line $\operatorname{Re} s = \sigma$ as t moves along the contour, and the configuration at $t = \tau$ persists. In the stationary cases the path is (γ_0, W) with γ_0 an independent Gaussian coordinate, and (4) is again a quadratic form of rank two, in $(J(0), J(1))$ in case SK; the argument is unchanged. With a fixed start the same argument, with the mean of ξ carried along, shows that the weights do not depend on the start and that the configuration of the roots is the same. □

Proof of Theorem 2 and Corollary 1. Theorem 1 is applied to $(X,Y) = (U,V)$, on the tube $\operatorname{Re} s = \sigma$, $\operatorname{Re} t = \tau$ of Lemma 1, through the smoothing in its proof rather than through hypothesis (19), which fails in two of the configurations the theorem covers: at $q = 0$ when $c_0 = 0$, where $J_2(0,t)$ does not depend on s and $|M(s,t)|$ decays only like $|s|^{-r/2}$; and in case DK when $c_0 \neq 0$ and the parabola $X(v)$ has its vertex at a positive v, that is $c_0 x < 0$, where $2J_2(xs, t + c_0 s)$, by stationary phase at $v = -x/(2c_0)$, decays only like $|s|^{-1/2}$, as does $|M|$. Hypothesis (18) holds: $|M(s,t)| \leq C|P(t)|(1 + |s|)^{-r/2}$ on the tube, because the roots $s_j(t)$, which are of the order of $|t|^{1/2}$ for large $|t|$, stay off the line $\operatorname{Re} s = \sigma$, and $|P(t)| \leq C\exp(-\frac{1}{2}|\operatorname{Im} t|^{1/2})$. So, for $\epsilon > 0$, $M(s,t)e^{\epsilon^2 s^2/2}$ satisfies (A.1), and by the first

part of the proof of Theorem 1 the variable $Q_\epsilon = (U + \epsilon Z - c_0 V)/\sqrt{V}$ has the continuous density f_ϵ given by (20) with $e^{\epsilon^2 s^2/2}$ inserted. In that absolutely convergent integral the s- and v-integrals may be interchanged, by Fubini's theorem, once J_2 is written out:

$$\frac{1}{2\pi \mathrm{i}} \int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} M(s,t) e^{\epsilon^2 s^2/2}\, 2J_2(xs, t + c_0 s)\mathrm{d}s$$
$$= \int_0^\infty 2v^2 e^{-tv^2} \Big\{ \frac{1}{2\pi \mathrm{i}} \int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} M(s,t) e^{\epsilon^2 s^2/2} e^{-s(xv + c_0 v^2)} \mathrm{d}s \Big\} \mathrm{d}v,$$

and the inner integral is $P(t)\varphi_{t,\epsilon}(X(v))$ with $\varphi_{t,\epsilon} = \varphi_t * g_\epsilon$, the convolution of (23) with the $N(0, \epsilon^2)$ density g_ϵ, since $e^{\epsilon^2 s^2/2}$ is the transform of g_ϵ and (22) that of φ_t. Hence

$$f_\epsilon(x) = \frac{1}{2\pi \mathrm{i}} \int_{\tau-\mathrm{i}\infty}^{\tau+\mathrm{i}\infty} P(t)\, \Phi^f_{x,\epsilon}(t) \mathrm{d}t,$$

with $\Phi^f_{x,\epsilon}$ the first transform of (26) with $\varphi_{t,\epsilon}$ in place of φ_t. Now let $\epsilon \to 0$. Off $X = 0$ the function φ_t is continuous, so $\varphi_{t,\epsilon}(X) \to \varphi_t(X)$ for every $X \neq 0$. On the tube $|\varphi_t(X)| \leq C(t) e^{-\sigma X} h(X)$, with $h(X) = X^{-1/2} \mathbf{1}\{X > 0\}$ in case DK and $h(X) = 1 + |\log |X||$ otherwise, since $\operatorname{Re} s_2 < \sigma < \operatorname{Re} s_1$ and $|K_0(z)| \leq C(1 + |\log |z||) e^{-\operatorname{Re} z}$ for $\operatorname{Re} z \geq 0$, with $C(t)$ bounded by a power of $|t|$; and for $\epsilon \leq 1$ the smoothed function obeys the same bound with $h + 1$ in place of h, because the convolution of h with a positive radially decreasing kernel of unit mass is at most the Hardy–Littlewood maximal function of h, which for these two h is a constant multiple of h, and the factor $e^{\sigma y}$ that the weight $e^{-\sigma X}$ leaves under the convolution is harmless when $\epsilon \leq 1$. The dominating function $2v^2 e^{-\tau v^2} e^{-\sigma X(v)} \{h(X(v)) + 1\}$ is integrable over $v > 0$, as $\tau + c_0 \sigma > 0$ and the zeros of $X(v)$ are simple, except in case DK with $c_0 > 0$ at $x = -\sqrt{2c_0}$, where X has a double root and $h(X(v))$ is not integrable, which is the point at which the density is infinite. By dominated convergence $\Phi^f_{x,\epsilon}(t) \to \Phi^f_x(t)$, and then, as $|P(t)|C(t)$ is integrable along the line, $f_\epsilon(x)$ converges to the right side of (25), uniformly on compact sets of x that avoid $-\sqrt{2c_0}$ in that case, and the limit is continuous there by the same domination. The step in italics in the proof of Theorem 1 then shows that the right side of (25) is the density of the limit and that its distribution function is continuous, on $(-\infty, -\sqrt{2c_0})$ and $(-\sqrt{2c_0}, \infty)$ in that case, hence also at $-\sqrt{2c_0}$, where the singularity of the density is logarithmic and integrable, and on $\mathbb{R}$ otherwise. For (21) the same steps with

$$\frac{2J_1(xs, t + c_0 s)}{s} = 2 \int_0^\infty v e^{-tv^2} \frac{e^{-s(xv + c_0 v^2)}}{s} \mathrm{d}v, \qquad \frac{e^{-su}}{s} = \int_u^\infty e^{-sy} \mathrm{d}y,$$

give $\mathbb{P}(Q_\epsilon > x)$ as the integral over v of $2tve^{-tv^2}$ times $P(t)\Psi_{t,\epsilon}(X(v))$, divided by t, with $\Psi_{t,\epsilon} = \Psi_t * g_\epsilon$, and as $\epsilon \to 0$ this converges to the first form of Φ_x in (26), Ψ_t being continuous and bounded by $C(t)e^{-\sigma X}$ on the tube, while $\mathbb{P}(Q_\epsilon > x) \to \mathbb{P}(\tau > x)$ at every x because the distribution function is continuous; the $1/t$ comes from writing $2ve^{-tv^2}$ as $2tve^{-tv^2}/t$. The second form is the integration by parts

$$\int_0^\infty \frac{\mathrm{d}}{\mathrm{d}v}\big(-e^{-tv^2}\big)\Psi_t\big(X(v)\big)\mathrm{d}v = \Psi_t\big(\tfrac{1}{2}\big) + \int_0^\infty e^{-tv^2}X'(v)\varphi_t\big(X(v)\big)\mathrm{d}v.$$

At $c_0 = 0$ the closed forms of Corollary 1 follow by writing the integrals over $(0, \frac{1}{2})$ as the difference of the integrals over $(0,\infty)$ and $(\frac{1}{2},\infty)$ in case DK, where $\int_0^\infty X^{k-1/2}e^{-\beta X^2-\gamma X}\mathrm{d}X$ is $\mathcal{P}_k$ and $\int_{1/2}^\infty$ is $\mathcal{Q}_k$ after $X = \frac{1}{2}(1+u)$, and by the definition (28) otherwise. □

Lemma 2 (The tilted process given its start)**.** *In case* DK *with a fixed start* γ_0*, for real* $t < t_*$ *and real* u*,*

$$\mathbb{E}\big\{e^{\mathrm{i}uJ(1)+tV}\big\} = \{e^cF_0(z)\}^{-1/2}\exp\Big(-\frac{u^2}{2}\frac{F_1(z)}{F_0(z)} + \mathrm{i}u\frac{\gamma_0}{F_0(z)} + \gamma_0^2\frac{tF_1(z)}{F_0(z)}\Big), \tag{A.2}$$

with F_0*,* F_1 *of case* DK *at* $z = 2t - c^2$*. Under the tilt* e^{tV}*,* $J(1)$ *given* $J(0) = \gamma_0$ *is therefore Gaussian with mean* γ_0/F_0 *and variance* F_1/F_0*, and the tilt multiplies the moment generating function of* V *by* $e^{\gamma_0^2 tF_1/F_0}$*.*

Proof. Let $h(r,x) = \mathbb{E}\{\exp(\mathrm{i}uJ(1) + t\int_r^1 J(q)^2\mathrm{d}q) \mid J(r) = x\}$. By the Feynman–Kac formula, $\partial_r h + cx\,\partial_x h + \frac{1}{2}\partial_{xx}h + tx^2h = 0$ with $h(1,x) = e^{\mathrm{i}ux}$. The trial $h = \exp\{\alpha(r)x^2 + \beta(r)x + \delta(r)\}$ gives the Riccati system $\alpha' + 2c\alpha + 2\alpha^2 + t = 0$, $\beta' + (c+2\alpha)\beta = 0$, $\delta' + \alpha + \frac{1}{2}\beta^2 = 0$ with $\alpha(1) = 0$, $\beta(1) = \mathrm{i}u$, $\delta(1) = 0$. With $\alpha = k'/(2k)$ the first equation is $k'' + 2ck' + 2tk = 0$, solved by $k(r) = e^{-cr}g(r)$, $g(r) = \cosh\mu(1-r) - (c/\mu)\sinh\mu(1-r)$, $\mu^2 = c^2 - 2t = -z$, which satisfies $k'(1) = 0$; then $g(0) = C - cS = F_0$, $k'(0)/k(0) = 2tS/F_0$ and $\alpha(0) = tF_1/F_0$. The second gives $\beta = \mathrm{i}ue^{-cr}/k(r)$, so $\beta(0) = \mathrm{i}u/F_0$. The third gives $\delta(0) = \int_0^1\{k'/(2k) + \frac{1}{2}\beta^2\}\mathrm{d}r = -\frac{1}{2}\log\{e^cF_0\} - \frac{1}{2}u^2\int_0^1 g(r)^{-2}\mathrm{d}r$, and since $g'' = \mu^2 g$, the Wronskian of g and $g_2(r) = \sinh\mu(1-r)/\mu$ is constant, $gg_2' - g'g_2 = -1$, so that $\int_0^1 g^{-2}\mathrm{d}r = g_2(0)/g(0) = S/F_0$. Evaluating h at $(0,\gamma_0)$ gives (A.2); its continuation to complex t and u holds on the tube. □

Proof of Proposition 2. In case DK, $R = \frac{1}{2}\{J(1)^2 - \gamma_0^2\}$, and by Lemma 2, under the tilt, $J(1) \sim N(\gamma_0/F_0, F_1/F_0)$ while M_V gains $e^{\gamma_0^2 tF_1/F_0}$; the moment generating function of $\frac{1}{2}X^2$

for $X \sim N(m,w)$ is $(1-sw)^{-1/2}\exp\{\frac{1}{2}sm^2/(1-sw)\}$, which gives (9) after the subtraction of the zero-start value. In cases DU and DT the dependence (7)–(8) of $\log M$ on the start is Theorems 1 and 2 of Nabeya and Sørensen (1994), in the notation of Proposition 1, whose parameter f is $c\gamma_0$ and $c^2\gamma_0$ here. For the partial fractions, note that R is invariant to $J \mapsto J+a+br$ (to $J \mapsto J+a$ in case DU), as the regression is; one checks this directly on (4). The functionals $(J(0), J(1), \bar{J}, K)$ change by $(a, a+b, a+b/2, a/2+b/3)$, so R is a quadratic form in two linear combinations of them that are invariant, whose coefficients do not involve the start. With $J = \gamma_0 h + J_0$, $h(r) = e^{cr}$ and J_0 the zero-start process, these combinations are Gaussian with means proportional to γ_0, also under the tilt, and the covariance under the tilt does not involve γ_0; so $R = \frac{1}{2}(X_1^2 - X_2^2)$ with X_1, X_2 independent Gaussian in the eigenbasis of the quadratic form, of variances $2\kappa_1$, $-2\kappa_2$ with the weights of the zero start, and means proportional to γ_0. The moment generating function of this noncentral form is $\prod_{k=1}^2(1-2s\kappa_k)^{-1/2}\exp\{\frac{s}{2}\sum_{k=1}^2(-1)^{k-1}m_k^2/(1-2s\kappa_k)\}$, so that $Q(s,t) - Q(0,t)$ is of the form (10) with $m_k^2 = \gamma_0^2\mu_k$, and the μ_k are read off the residues of (8) at the roots: with $D(s,t) = e^c F_2(s-s_1)(s-s_2)$ the residue of Q at s_k is $c^{2k}(t+\frac{1}{2}s_k^2 + cs_k)g(z)/\{F_2(s_k - s_{k'})\}$, $k' \neq k$, and that of the right side of (10) is $-\frac{1}{2}\mu_1 s_1^2$ at s_1 and $\frac{1}{2}\mu_2 s_2^2$ at s_2. Both sides are rational in s with the same poles and the same value at $s=0$, and tend to constants as $s \to \infty$, so they agree once the residues do. □

Proof of (11) and (12). Since $\gamma_0^2 \sim \sigma_0^2\chi_1^2$ with $\sigma_0^2 = -1/(2c)$, $\mathbb{E}\exp(\gamma_0^2 Q) = (1-2\sigma_0^2 Q)^{-1/2} = (1+Q/c)^{-1/2}$, and $D(1+Q/c)$ is (11). In case SK, by Lemma 2, averaging over $\gamma_0 \sim N(0,\sigma_0^2)$ with the weight $e^{\gamma_0^2 tF_1/F_0}$ leaves $J(0)$ Gaussian with variance $\tilde{\sigma}^2 = \sigma_0^2/(1-2\sigma_0^2 tF_1/F_0)$ and multiplies the moment generating function of V by $(1-2\sigma_0^2 tF_1/F_0)^{-1/2}$, so that under the tilt the pair $(J(0), J(1))$ is Gaussian with $\operatorname{Var} J(0) = \tilde{\sigma}^2$, $\operatorname{Cov}\{J(0), J(1)\} = \tilde{\sigma}^2/F_0$ and $\operatorname{Var} J(1) = \tilde{\sigma}^2/F_0^2 + F_1/F_0$. With $R = \frac{1}{2}\{J(1)^2 - J(0)^2\}$ and $\Lambda = \operatorname{diag}(-1,1)$, $\mathbb{E}e^{sR} = \det(I - s\Sigma\Lambda)^{-1/2}$, so $D_{\mathrm{S}} = e^c F_0(1-2\sigma_0^2 tF_1/F_0)\{1 - s(\Sigma_{22}-\Sigma_{11}) - s^2\det\Sigma\}$. Here $\det\Sigma = \tilde{\sigma}^2 F_1/F_0$, and $\Sigma_{22}-\Sigma_{11}$ vanishes: by $C^2 - zS^2 = 1$, $1-2tS^2 = (C-cS)(C+cS) = F_0(C+cS)$, so that $\tilde{\sigma}^2/F_0^2 + F_1/F_0 - \tilde{\sigma}^2$, which equals $\{\sigma_0^2(C+cS) + F_1 - \sigma_0^2 F_0\}\tilde{\sigma}^2/(\sigma_0^2 F_0)$ after the substitution of $\tilde{\sigma}^2$, is $(2c\sigma_0^2 S + S)\tilde{\sigma}^2/(\sigma_0^2 F_0) = 0$ since $2c\sigma_0^2 = -1$. Multiplying out gives (12). □

Proof of Theorem 3. By (4), R is a quadratic form $R = \frac{1}{2}\boldsymbol{\ell}'\boldsymbol{Q}\boldsymbol{\ell}$ in a vector $\boldsymbol{\ell}$ of at most three Gaussian functionals of W, of rank one or two. Approximate W on a grid of N points: V becomes a quadratic form $\boldsymbol{x}'\boldsymbol{K}_V\boldsymbol{x}$ in $\boldsymbol{x} \sim N(\boldsymbol{0}, \boldsymbol{I}_N)$, with $\boldsymbol{K}_V$ positive semidefinite, and $\boldsymbol{\ell} = \boldsymbol{\Gamma}\boldsymbol{x}$. Let $t = \sigma + \mathrm{i}y$ with σ real and $\boldsymbol{P} = \boldsymbol{I}_N - 2t\boldsymbol{K}_V$, whose real part $\boldsymbol{I}_N - 2\sigma\boldsymbol{K}_V$ is positive definite when $\sigma < t_*$. For bounded measurable h, completing the square in the Gaussian integral gives

$$\mathbb{E}\{e^{tV}h(\boldsymbol{\ell})\} = |\boldsymbol{P}|^{-1/2}\int h(\boldsymbol{l})\,\phi_{\boldsymbol{\Sigma}}(\boldsymbol{l})\mathrm{d}\boldsymbol{l}, \qquad \boldsymbol{\Sigma} = \boldsymbol{\Gamma}\boldsymbol{P}^{-1}\boldsymbol{\Gamma}',$$

where $\phi_{\boldsymbol{\Sigma}}$ is the Gaussian density formula evaluated at the complex covariance $\boldsymbol{\Sigma}$, a complex-valued function. It is integrable because $\operatorname{Re}\boldsymbol{\Sigma}^{-1}$, a Schur complement of the accretive matrix $\boldsymbol{P}$ after a change of coordinates, is positive definite. With $h = \mathbf{1}\{R \le a\}$ the integral is an analytic function of $\boldsymbol{\Sigma}$ on this set. For real $\boldsymbol{\Sigma}$ it is the probability that $\kappa_1 Z_1^2 + \kappa_2 Z_2^2 \le a$, where κ_1, κ_2 are the nonzero eigenvalues of $\frac{1}{2}\boldsymbol{\Sigma}\boldsymbol{Q}$, which equals (37), as the integral over Z_1 given Z_2 shows. The right side of (37) is analytic in (κ_1, κ_2) where $\operatorname{Re}\kappa_1 > 0 > \operatorname{Re}\kappa_2$, so the two agree there by analytic continuation. The moment generating function of R under the tilt, $\prod_{k=1}^{2}(1 - 2s\kappa_k)^{-1/2}$, identifies the κ_k with $1/(2s_k)$, by Lemma 1. Letting $N \to \infty$, the finite-dimensional quantities converge to those of the limit, locally uniformly in t, because the moment generating functions do. Finally $\mathbb{E}\{e^{tV}\mathbf{1}(R \le a)\}$ is the Laplace transform of the measure $\mathbb{E}\{\,\cdot\,; R \le a\}$ in v, and (36) is its inversion on the line $\operatorname{Re} t = \sigma$. For the path Γ, the integrand of (36) is analytic on the region between Γ and the line: the singularities of M_V are the zeros of $D(0, \cdot)$, the Fredholm determinant of the positive semidefinite covariance kernel of V, which are real; the weights are analytic wherever they are distinct, which the sign conditions ensure, and $\mathcal{P}$ is analytic in them there. By assumption, the integrals over the arcs of the circles $|t| = \varrho$ that join Γ and the line tend to zero as $\varrho \to \infty$; Cauchy's theorem on the bounded regions cut off by these arcs then moves the line to Γ.

For part (ii), take h in the Gaussian integral to be a function of R alone. For real $\boldsymbol{\Sigma}$ the law of R is that of $\kappa_1 Z_1^2 + \kappa_2 Z_2^2$, whose density is $(-s_1 s_2)^{1/2}\varphi_t$ (respectively $s_1^{1/2}\varphi_t$): in case DK the scaled χ_1^2 density, and in cases DU and DT the convolution of the densities of $\kappa_1 Z_1^2$ and $\kappa_2 Z_2^2$, which reduces with $K_0(w) = \int_0^\infty e^{-w\cosh q}\mathrm{d}q$ to (23). Both sides are analytic in (κ_1, κ_2) where $\operatorname{Re}\kappa_1 > 0 > \operatorname{Re}\kappa_2$, so, as for part (i), $\mathbb{E}\{e^{tV}h(R)\} = P(t)\int_{-\infty}^{\infty} h(r)\varphi_t(r)\mathrm{d}r$ on the line, and in

the limit $N \to \infty$. Hence $P(t)\varphi_t(r)$ is the Laplace transform in v of the joint density of (R, V), and (35) is its inversion. The integral converges absolutely for $r \neq 0$: with $\operatorname{Re} s_1 > 0 > \operatorname{Re} s_2$, $|e^{-(s_1+s_2)r/2} K_0((s_1 - s_2)|r|/2)|$ behaves like $|e^{-s_1 r}|$ for $r > 0$ and like $|e^{-s_2 r}|$ for $r < 0$ when $|t|$ is large, so $\varphi_t(r)$ grows at most like a power of $|t|$, while P decays like $\exp(-\frac{1}{2}|\operatorname{Im} t|^{1/2})$. Then $H(v; x) = \int_{-\infty}^{a(v)} h(r, v)\mathrm{d}r$, and differentiating $\int_0^\infty H(v; x)\mathrm{d}v$ in x, with $\partial a/\partial x = \sqrt{v}$, gives (38). For the distribution function, $G(r, v) = \int_v^\infty h(r, v')\mathrm{d}v'$ is the density of R on the event $\{V > v\}$, and for $\sigma' > 0$ the integral of (35) over $v' > v$ may be taken under the t-integral, which gives (40). Moving its line across the simple pole at $t = 0$, where $M_V(0) = 1$, gives the residue $(-s_1 s_2)^{1/2}\varphi_0(r)$. With $\Psi(v) = \int_{-\infty}^{a(v)} G(r, v)\mathrm{d}r = \mathbb{P}\{R \le a(v),\, V > v\}$ and $\partial G/\partial v = -h$,

$$\Psi'(v) = a'(v)\, G\big(a(v), v\big) - H(v; x).$$

Since $\Psi(0) = \mathbb{P}(R \le \frac{1}{2})$, $\Psi(v) \to 0$ as $v \to \infty$, and $G(r, v)$ is bounded near $v = 0$ for $r = a(v)$ near $\frac{1}{2}$, so that $a'(v)G(a(v), v)$ is integrable at $v = 0$, integrating over $v \in (0, \infty)$ gives (39). At $t = 0$ the weights are real, and $\mathbb{P}(R \le \frac{1}{2}) = \mathcal{P}(\kappa_1(0), \kappa_2(0); \frac{1}{2})$ by part (i). With a fixed start the Gaussian integral has the deterministic part of $\boldsymbol{\ell}$ carried along, $\mathbb{E}\{e^{tV} h(\boldsymbol{\ell})\} = |\boldsymbol{P}|^{-1/2} e^{\gamma_0^2 Q(0,t)} \int h(\boldsymbol{l})\phi_{\boldsymbol{\Sigma}}(\boldsymbol{l} - \boldsymbol{m})\mathrm{d}\boldsymbol{l}$ with a complex mean $\boldsymbol{m}$ proportional to γ_0, and for real $\boldsymbol{\Sigma}$ and $\boldsymbol{m}$ the integral with $h = \mathbf{1}\{R \le a\}$ is $\mathbb{P}\{\frac{1}{2}(X_1^2 - X_2^2) \le a\}$ for independent $X_k \sim N(m_k, \pm 2\kappa_k)$, which is (41) after conditioning on X_2 and replacing X_2 by $|X_2|$; the right side of (41) depends on the means only through m_k^2 and is analytic in the weights and in m_k^2, so analytic continuation and the limit $N \to \infty$ give the result as before, with $m_k^2 = \gamma_0^2 \mu_k$ by Proposition 2. With a stationary start R is a central form of rank two in the Gaussian vector that includes γ_0, and the theorem applies with D_{S}. □

Appendix B. Numerical details

Appendix B.1. The single integral

The transforms. The transforms (26) are integrals of φ_t against Gaussian kernels over half-lines and finite segments of the v-axis. At $c_0 = 0$ in case DK the segment integrals are written as differences of half-line integrals, which are the special functions (27); otherwise the Gaussian transforms are evaluated by quadrature as follows, and the same rules, with

the K_0 kernel replaced by $X^{-1/2}$, evaluate (27). For $x < 0$ and $c_0 = 0$ the integrals over $(0, \frac{1}{2})$ in X contain the contributions of the support edge $X = 0$, which carry the factor e^{-ty}, $y = 1/(4x^2)$, and oscillate in $\operatorname{Im} t$ with the period $8\pi x^2$, and the contributions of the neighborhood of $X = \frac{1}{2}$, where the Gaussian factor is centered, which are analytic in t without oscillation. They are separated by a partition of unity, $\chi(X) = \frac{1}{2}[1 + \operatorname{erf}\{\kappa(\frac{1}{4} - X)\}]$ with $\kappa = 24$, applied to the integrand over $(0, \frac{1}{2})$: χ is entire, 1 and 0 at the ends of the segment to 10^{-16}, so that both parts are integrals of analytic functions between the same real endpoints, hence analytic and conjugate symmetric in t, and only the part with χ carries the factor e^{-ty}. The partition is applied to the complex path point on a fixed path, so that its definition does not depend on t. In case DK the same partition, in the variable v with $\kappa = 12/v^*$, $v^* = 1/(2|x|)$, reproduces the split of (29) up to terms of relative size 10^{-16}.

The paths. The half-lines from $X = \frac{1}{2}$ and from the edge $X = 0$ outward are rotated by $-\frac{1}{2}\arg t$, on which the Gaussian factor decays without oscillating, and the segment $(0, \frac{1}{2})$ is bulged into the half-plane $\operatorname{Im}(X - \frac{1}{2})^2 < 0$ along a fixed curve with unit slope at both ends, the direction of steepest descent there, and height $\frac{1}{16}$, which keeps $|\chi| \le e^{9/4}$ on it. The endpoint singularities, $X^{-1/2}$ in case DK and $\log X$ otherwise, are removed by the substitution $X = \delta u^4$; each path is divided at the scales of its factors, the width $|x|(2 \operatorname{Im} t)^{-1/2}$ of the Gaussian, the decay lengths of the exponentials and the distance $1/\kappa$ over which χ varies, with 32 to 96 nodes per piece and the pieces growing geometrically along the half-lines. The constant part of the kernel in the cases with the K_0 kernel is the total mass $(-s_1 s_2)^{-1/2}$ minus a short integral over $X > \frac{1}{2}$ on the ray $X = \frac{1}{2} + re^{-\mathrm{i}\arg s_1}$, on which φ_t decays without oscillating, or, when $|s_1|$ is small, the integral over $X < \frac{1}{2}$ with the ray $X = -re^{-\mathrm{i}\arg(-s_2)}$ from the edge. The branches of $P(t)$, $s_1(t)$, $s_2(t)$ and $(-s_1 s_2)^{-1/2}$ are fixed by continuity from the real point $t = \tau$ along each set of nodes; for $c > 2$ the roots and $P(t)$ come from the Girsanov forms (15), and the smaller root in modulus from the product of the roots rather than the quadratic formula, which loses it to cancellation when the roots differ by orders of magnitude (at $c = 10$ they are about -100 and 3×10^{-8}).

The Bessel function. The function $e^z K_0(z)$, needed at points with $\operatorname{Re} z \ge 0$ only, is evaluated to 10^{-14} by its power series for $|z| < 2$, by Taylor expansions about the centers of cells of

side $\frac{1}{2}$ for $|z| < 25$, the coefficients following from K_0 and K_1 at the center through Bessel's equation, and by its asymptotic series beyond, at a tenth of the cost of a library routine.

The outer integral. The part without oscillating factors is integrated by the trapezoidal rule in u, $w = \tau \sinh u$, with the step 0.075 up to $w = 4900$ (123 nodes). Each part with a factor $e^{-tv_j^2}$ is integrated by Filon's method on panels that grow geometrically from $\min(\frac{1}{2}, 1/v_j^2, \tau/2)$, the scale of the pole at $t = 0$ and of the transient of the transition region of χ, which oscillates at the frequency $3v_j^2/4$ and dies out over $300/v_j^2$; below $w = 300/v_j^2$ the panels are at most $25/v_j^2$ long, G_j is interpolated at 20 Gauss–Legendre nodes by Legendre polynomials, and the moments $\int_{-1}^{1} P_n(x) e^{-\mathrm{i}\Omega x} \mathrm{d}x = 2(-\mathrm{i})^n j_n(\Omega)$ integrate the oscillation exactly; 18 to 25 panels reach $w = 4900$. This part is omitted when $3\tau v_j^2/16 > 40$, where it is below 10^{-17}.

The saddle factor. For $c > 0$ the roots s_1, s_2 at $t = \tau$ approach zero as c grows, and the Gaussian parts of the transforms, of the form $\int_0^\infty A(u) e^{-tu^2/x^2 - su} \mathrm{d}u$ with A algebraic and $s = s_1$ or s_2, grow or oscillate along the ray $u = re^{-\mathrm{i}\arg(t)/2}$ for $x > 0$ large: the model integral with $A = 1$ is $\frac{1}{2} x (\pi/t)^{1/2} e^{z^2} \operatorname{erfc}(z)$ with $z = sx/(2t^{1/2})$, which for $\operatorname{Re} z < 0$ contains the term $2e^{z^2}$ of the saddle point $u_0 = -sx^2/(2t)$ of the exponent. For $\operatorname{Re} z < 0$ these parts are taken along the segment from 0 to u_0 and the line of steepest descent from u_0, on which the exponent is $z^2\lambda(2 - \lambda)$, $0 \le \lambda \le 1$, and $z^2 - |t| r^2/x^2$, and otherwise along the level curve $\operatorname{Im}(tu^2/x^2 + su) = 0$ from 0, on which the exponent is $-v^2$ for real v and $u = 2v^2/\{s + (s^2 + 4tv^2/x^2)^{1/2}\}$, which ends in the valley of the positive real axis precisely when $\operatorname{Re} z \ge 0$. The factor e^{z^2}, present for $c > 0$ beyond a small value of w and never for $c \le 0$, where $\operatorname{Re} z \ge 0$ along the line, oscillates in w with a phase that grows like x^2; where its change between the nodes of the trapezoidal rule exceeds $\frac{1}{4}$, the smooth part is integrated instead on the geometric panels with Gauss–Legendre rules whose node counts follow the phase. The cost then grows with x^2 in the upper tail.

The local null value. With $c_0 \ne 0$ the breakpoints of the inner integral in v are the positive roots of $X(v)$ and, for $c_0 > 0$ and $-\sqrt{2c_0} < x < 0$, the vertex $v_0 = -x/(2c_0)$ of the parabola, where X has the complex roots $v_0 \pm \mathrm{i}\ell$, $\ell = (X(v_0)/c_0)^{1/2}$, from which the cuts of $X^{-1/2}$

or $\log X$ run vertically away from the real axis; the paths leave v_0 within 45 degrees of the real axis, so that they pass the branch points at the distance 0.7ℓ at least, the first pieces of the quadrature at v_0 have the length ℓ, and the ray from v_0 is straight. The first piece at a root v_j has the length $\ell_j/2$, ℓ_j the distance of the nearest other root, and the first piece at $v = 0$ the length $\ell_0/2$, ℓ_0 the distance of the nearest root from 0 (for $c_0 < 0$ and large x the negative root $1/(2c_0v_1) \approx -1/(2x)$ lies close to 0). On a finite segment between two breakpoints $a < b$ the integrand is $W(v)e^{E(v)}\varphi_t^{\mathrm{alg}}(X(v))$ with the exponent $E(v) = -tv^2 - s_dX(v) = -\alpha v^2 - \beta_0 v - \frac{1}{2}s_d$, $\alpha = t + c_0s_d$, $\beta_0 = s_dx$, a quadratic in v, and the path is three straight pieces: from a and from b in the directions of steepest descent of E at the ends, each kept within 80 degrees of the segment and stopped where the exponent has fallen by 60, where its quadratic part turns it up, at the depth $L/8$, or at the half-length $L/2$; and the connection between their ends, which is the straight line unless the exponent along it rises above its values at the ends, and otherwise the polyline through the real axis, on which the exponent is that of the integrand itself (when $\operatorname{Re}\alpha \approx 0$ and $\operatorname{Im}\beta_0$ is large the two descent pieces end on opposite sides of the real axis, and a straight connection crosses the quadrant in which $e^{-\alpha v^2}$ grows by e^{17} and more). Every piece is integrated by Gauss–Legendre rules on sub-pieces graded toward the ends, with the node counts set by the change of E in modulus and phase at 1.2 nodes per unit plus 16 for a stationary point of the phase inside, and sub-pieces on which the exponent is below $\max\{E(a), E(b)\} - 58$ are skipped; X is formed from the offset from the nearer root, so that it is exact near the roots. The exponent has its saddle at $v^* = -\beta_0/(2\alpha)$, where it exceeds its value at 0 by $\beta_0^2/(4\alpha)$. With $c_0 < 0$ and large x the saddle sits inside the segment at $v^* \approx b/4$, with a phase of about $0.005x^2 \operatorname{Im} t$ in cases DU and DT at $c = 5$, which the trapezoidal rule cannot follow; the phase is therefore computed also when a breakpoint is present, whenever $0 < \operatorname{Re} v^* < v_1$, and the smooth part takes the Gauss–Legendre route of the saddle factor, with the partition χ between the smooth and the edge parts placed midway between v^* and v_1 and sharpened to $\kappa = 12/(v_1 - v^*)$, so that the saddle leaks into the edge part by at most $\operatorname{erf}(-6)$, and the depth of the paths reduced to $1.5/\kappa$. The saddle part needs about three outer nodes per radian of the phase over the range in which it is not negligible, about 600 at $x = 57$ and

3000 at $x = 266$ for $c = 5$, $c_0 = -5$, in addition to the 400 or so of the other parts. For $c_0 = 0$ the general machinery reproduces the values of the specialized code to 10^{-13}.

The fixed start. The paths and partitions are those of the local null value, with the constant $\frac{1}{2}$ of the parabola replaced by $\frac{1}{2}(1+\gamma_0^2)$ in case DK, and with the exponent that places the nodes, $-tv^2 - s_d X$, augmented by $\log\cosh\{b_2(2\tilde{X})^{1/2}\}$, the growth of (32) and (33) away from the edge, whose derivative also enters the descent directions; the factor carried by the nodes is φ_t divided by that growth, which is bounded. Laid out from the first two terms alone, the paths miss the mass of φ_t beyond the breakpoint, and the result depends on τ at the level of 10^{-11}. The half-line beyond the last breakpoint is the ray in the direction of the Gaussian's decay, with the pieces and node counts from the full exponent, or, when the exponential grows along it by more than e^2, the segment to the saddle of the quadratic exponent and the descent line from it. The integral G of (33) is a Gauss–Legendre rule in T after $l = |2Y|^{1/2}\sinh T$, with 40 nodes up to where the Gaussian, against the growth of the hyperbolic cosines, is below e^{-41}; 40 nodes agree with 384 to 10^{-14}. The Gaussian times the growth $e^{b_1 l}$ peaks at $l^* = b_1/\operatorname{Re} a$ with the width $(\operatorname{Re} a)^{-1/2}$, and when that peak is separated from the scale $|2Y|^{1/2}$ of the edge, at large noncentrality, the rule in T stops at $l^* - 8(\operatorname{Re} a)^{-1/2}$ and a Gauss–Legendre rule in l, with the node count from the change of the phase, covers the peak; the terms are formed relative to the largest, $G = Se^E$, so that nothing overflows, and $\cosh z$ as $e^z(1+e^{-2z})/2$ with $\operatorname{Re} z \geq 0$. The rule runs along the ray $l = e^{-\mathrm{i}\arg(a)/2}s$, on which the Gaussian of G is real, so that its branch cut in Y lies where aY is negative real, as the cut of K_0 in (23) does, and the paths, which avoid that cut, serve. The same mass that moves the peak of G moves the mass of φ_t beyond the breakpoint: on the ray in the direction of the Gaussian's decay the integrand there oscillates with a magnitude of up to e^λ around a value of order one, so when the full exponent rises along the real axis by more than 2 the ray is replaced by the path through the saddle of the full exponent (Newton from the maximum on the real axis, itself refined by Newton), continued by straight pieces in the direction of steepest descent, each renewed where the descent along it ends; the saddle is used only when it is that maximum slightly displaced, within one unit in height and with an imaginary part at most half its real part, since a saddle elsewhere is

not one the real-axis path crosses: otherwise the ray is kept for a rise below 10, where its cancellation is harmless, and the real-axis maximum is the pivot of the path beyond. With $c > 0$ the initial-value term $e^{\gamma_0^2 Q(0,t)}$ has an essential singularity at t_*, and the line stays at $\tau \le t_*/2$, where $\tau = t_*/4$ gives the same values to 10^{-15}. In the C implementation the rule for G is written in real arithmetic on arrays, with exponentials and sines from branch-free polynomial routines that the compiler vectorizes; a value of the distribution function then takes 7 to 80 milliseconds on ten threads of a laptop at $\lambda \le 12$, against 0.25 seconds by the double integral, and up to 0.3 seconds at $\lambda = 200$.

The moments. The derivatives in (43) are Cauchy integrals,

$$m_p(s) = \frac{p!}{N r^p} \sum_{k=0}^{N-1} M(re^{\mathrm{i}\theta_k}, -s - c_0 re^{\mathrm{i}\theta_k})\, e^{-\mathrm{i}p\theta_k}, \qquad \theta_k = \frac{2\pi k}{N},$$

with $N = 64$ and $r = 0.4 \min_k |s_k(-s)|$, the roots of $D(\cdot, -s)$ being real; a circle on which $\log D$ does not return to its starting value encloses a zero of D, and its radius is halved. The s-integral is the trapezoidal rule in $\log s$: the integrand is smooth in $\log s$ and decays at both ends, like $s^{p/2}$ at the left and like $\exp(-s^{1/2})$ at the right, so the rule converges geometrically.

Appendix B.2. The double integral

Zero start: the inversions. For each pair $(v, a(v))$ let $\ell(t)$ be the logarithm of the integrand, (42) in case DK and $\log\{P(t)\varphi_t(a(v))\} - tv$ in cases DU and DT, in the Girsanov forms of Section 3.1, with the exponent $-a\eta - tv$ of (42) written as $-(a - cv)\eta - cv\epsilon + \delta^2 v/2$, $\delta = \vartheta - c$ and $\epsilon = \eta - \delta$, which avoids the cancellation between $a\eta$ and tv at large v. We find its saddle point $\hat{s}$ on the real axis, left of the first point where one of $\cosh\vartheta\,|\boldsymbol{\Sigma}|$, α_0 and $\alpha_1^2 + 4\alpha_0\alpha_2$ leaves $(0, \infty)$ (found by bisection), by bracketing and the Illinois variant of regula falsi on ℓ', with finite-difference steps scaled by the width $\ell''(\hat{s})^{-1/2}$, and integrate along two paths through $\hat{s}$, keeping the result with the smaller error estimate. The first is

$$t(\tau) = \hat{s} + \mu \frac{\tau^2}{(1 + \tau^2/16)^{1/2}} - 2\mathrm{i}\mu\tau, \qquad \mu = \frac{1}{2\ell''(\hat{s})^{1/2}},$$

a parabola near $\hat{s}$ that turns into the wedge $\operatorname{Re} t - \hat{s} = 2|\operatorname{Im} t|$ far out. The second is the vertical line $\vartheta = \vartheta_A + \mathrm{i}y$ in the ϑ-plane, a parabola $t = (c^2 - \vartheta^2)/2$ opening to the right,

along which $|e^{-tv}|$ decays like $e^{-y^2v/2}$; we take $\vartheta_A = \max\{\hat{\vartheta}, \frac{1}{2}\operatorname{asinh}(a/(6v)^{1/2}), 1\}$, with $\hat{\vartheta}$ the value at $\hat{s}$, so that the oscillating part $a\vartheta(\coth\vartheta - 1)$ of $a\eta$ stays small against $y^2v/2$. In case DK a third route writes erf $= 1 -$ erfc and inverts the density of V and the remainder separately, each along a path through its own saddle point left of t_*, with erfc from the Faddeeva function (Weideman, 1994); it is the better one when $\mathcal{P}$ is close to one. Each integral uses the trapezoidal rule with 96 nodes after a sinh substitution, with the nested rule of twice the step as error estimate, and the node count is doubled up to three times until that estimate is below $10^{-13}\max(1, |I|)$, I the value. For G in (40) the crossing point is kept at least one saddle-point width from the pole at $t = 0$, and the residue is added when it lies to the left. The distribution function in case DK comes from part (i) and its density from (38), whose integrand $e^{-c/2}(\vartheta/\sinh\vartheta)^{1/2}e^{-a\eta}(\pi a)^{-1/2}$ is the derivative of (42) in a; in cases DU and DT both come from part (ii).

The outer integral. The outer integral over v uses Gauss–Legendre rules in $\log v$ on the interval from v_0 to $e^6\mathbb{E}_{\mathsf{DK}}V$, where

$$\mathbb{E}_{\mathsf{DK}}V = \frac{e^{2c} - 1 - 2c}{4c^2},$$

the mean of $\int_0^1 J(r)^2\mathrm{d}r$, bounds $\mathbb{E}V$ in all three cases, and $v_0 = 3\times10^{-3}$ in cases DU and DT and the smaller of 3×10^{-3} and $e^{-8}\mathbb{E}V$ in case DK; lowering v_0 to 10^{-5} changes the results by less than 10^{-10} for c down to -50. The parts are of length at most two, with 32 nodes in case DK and 64 in cases DU and DT, and with breakpoints at the zeros of $a(v)$, where the variable is mapped by $z \mapsto \frac{1}{2}(1 - \cos\pi z)$ (twice in cases DU and DT, for the logarithmic singularity of φ_t at $r = 0$), so that the distance to the breakpoint behaves like the fourth power of the variable. For $c_0 > 0$ and x near $-(2c_0)^{1/2}$ the two zeros of $a(v)$ merge, and the parts are graded geometrically toward them; at $x = -(2c_0)^{1/2}$ the density in case DK is infinite and is returned as such. Two further devices handle the far upper tail for $c > c_0$. There R is close to $(c - c_0)V$, and the outer integrand has a sharp step in $\log v$, and the density a narrow bump, near the point where $a(v)$ crosses the conditional mean of R given $V = v$. We compute that point and the width of the step from the tilt at the saddle point of $\log M_V(s) - sv$: the width is the conditional standard deviation of R, from the double

saddle-point formula $K_{uu} - K_{us}^2/K_{ss}$ for the conditional variance, divided by the slope of $a(v)$ minus the conditional mean, and when it is below 0.1 the parts are graded geometrically toward the point, starting from twice the width. In addition, a part is bisected, up to twelve times, when an error estimate from the Legendre coefficients of its integrand exceeds 10^{-12} and ten times the noise level of the inner integrals; the estimate is the larger of the last two coefficients, extrapolated by their decay from the middle ones. With the fixed rule alone, the distribution function in case DK at $c = 5$ is off by 3×10^{-3} at $x = 126$, with no warning, and the density is lost beyond $x \approx 320$. In (39) the boundary term becomes $\mathbb{P}\{R \le a(v_0)\}$ at the lower end v_0, with the weights at $t = 0$. Pairs $(v, a(v))$ at which the saddle-point approximation of the inner integral, times the outer weight, is below e^{-35} are skipped. The outer estimates, plus the outer rule applied to the error estimates of the inner integrals, are returned as an error estimate, and values whose estimate exceeds 10^{-10} are flagged.

With an initial observation. The inner integrals are one-dimensional inversions of $M_{\gamma_0}(0,t)\mathcal{P}$ (fixed start, through (41)) or of M_V times the tilted density (stationary start). On the line itself the integrand decays only like $\exp(-|y|^{1/2}/2)$ in $y = \operatorname{Im} t$, while e^{-tv} makes it oscillate with period $2\pi/v$, so a direct rule needs many nodes, and we use the parabola $t(\theta) = s^* + \mu\theta^2 - 2\mathrm{i}\mu\theta$, $\theta \in (-\infty, \infty)$, through the saddle point $s^*(v)$ of $\log M_V(s) - sv$ on the real axis (safeguarded Newton iteration, which keeps a bracket because $\log M_V$ is convex on its real domain), which opens to the right around the real singularities of M_V. Along it $|e^{-tv}| = e^{-s^*v}e^{-\mu v\theta^2}$. We take $\mu = \max\{1/(2\ell_2^{1/2}), 4.5\}$, ℓ_2 the second derivative of $\log M_V$ at s^*; the first choice turns the saddle-point Gaussian into $e^{-\theta^2/2}$, and the bound keeps $\operatorname{Re}(-z)^{1/2} \ge 3$ along the path, so that $e^{-2\vartheta}$ is small there. The sign conditions of Theorem 3 do not hold on the whole parabola: in case DK, with $w = z^{1/2} = u + \mathrm{i}v'$, $\kappa_1 = 1/\{2(w\cot w - c)\}$ and $\operatorname{Re}\kappa_1 > 0$ if and only if $u\sin 2u + v'\sinh 2v' > c(\cosh 2v' - \cos 2u)$, which fails near $u = 3\pi/4 + k\pi$ once u exceeds about $(2\mu)^{1/2}\sinh\{2(2\mu)^{1/2}\}$, at $|\theta| \approx 202$ for $c = 0$ and $\mu = 4.5$. We therefore use the parabola only for $|\theta| \le \Theta$, where $e^{-\mu v\Theta^2} < 10^{-18}$, and close the path by the vertical rays $\operatorname{Re} t = s^* + \mu\Theta^2$, whose contribution we neglect; the cutoff, like the range of the outer integral, is a practical choice, tested numerically: doubling the exponent at the cutoff and widening the range of v to $[e^{-12}, e^9]\mathbb{E}V$ changes

the distribution function by less than 10^{-11}. In all our computations $\Theta \le 9$, far inside the region where the conditions hold, and the routine checks them at every node it uses and on a grid of intermediate parabolas on request. Along the parabola $\log M_V = -\frac{1}{2}\log D(0,z)$ is continued from s^*, where $D > 0$: we write $\log D = c + \vartheta + \log q$ with $\vartheta = (-z)^{1/2}$, continuous on the path because $\mathrm{Im}(-z) > 0$ there, and $q = De^{-c-\vartheta}$, which varies slowly and whose logarithm is unwrapped on a grid refined until the phase changes by less than $\pi/4$ between neighbors. The real part of the integrand is even in θ, and the trapezoidal rule is applied to $\theta \ge 0$ after the substitution $\theta = \theta_w \sinh\tau$, with $\theta_w = 1/(2\mu\ell_2^{1/2})$ the width of the saddle-point Gaussian, with 96 nodes up to the point where $e^{-\mu v\theta^2} < 10^{-18}$, doubled up to twice where the nested-rule estimate exceeds 10^{-9} of the largest column of the piece; with a fixed start in cases DU and DT the initial-value term has an essential singularity at the end of the domain of $M_{\gamma_0}(0,\cdot)$, toward which s^* moves as v grows, and the crossing point is kept at a relative distance of 2% from the end. Only an approximate s^* is needed: any real point of the domain gives the same integral, and s^* and ℓ_2 only place and scale the path. The boundary term $\mathbb{P}(R \le \frac{1}{2})$ of (39) has real weights and is computed once for each x from (37) or (41), by 96-point Gauss–Legendre after the substitution $l = \sqrt{2a}\sinh\tau$ when $a > 0$ and $l = (u^2 - 2a)^{1/2}$ when $a < 0$, as is the inner integral over l of (41) at every node. The outer integral is the rule above in $\log v$ over $[e^{-8}, e^{6}]\mathbb{E}V$ with 64 nodes per part.

Accuracy of the double integral. Doubling the outer and the inner node counts changes the distribution function and the density by at most 5×10^{-11} with a zero start ($c \in [-30, 10]$, $c_0 \in [-40, 10]$) and by 1.5×10^{-9} at $c = c_0 = 20$. In the far upper tail at $c = 5$, where the outer integrand has the sharp step described above, the density integrated over $[-80, 500]$ by a 400-point Gauss–Legendre rule has mass $1 + 2 \times 10^{-9}$ in case DK and $1 + 2 \times 10^{-10}$ in case DU, and reproduces the means of Proposition 3 to 10^{-8}, the remaining differences being the truncation and the rule of that integral. With an initial observation, in case DK at $x = 0$, where the exact value is a noncentral χ^2_1 probability, the parabola route is accurate to 10^{-13} for $c \le 2$; in cases DU and DT with a fixed start it agrees with the limit of the Richardson-extrapolated finite-sample initial-value terms of Bao et al. (2026) to 10^{-7}, the accuracy of that extrapolation, and the sign conditions held at every node used for $c \le 2$.

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